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**SCHOOL OF ENGINEERING
AND COMPUTER SCIENCE**

**CareerFit: A Comparative and Explainable Resume-Job
Matching Framework for Application Readiness Assessment**
Final Report

Project and Report I - Summer 2026

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Executive Summary

CareerFit is a web-based academic prototype and research framework for explainable resume-job matching and application-readiness assessment. The project investigates how individual job requirements can be matched to specific resume evidence while maintaining transparency about why each matching decision is produced.

Rather than relying only on a single similarity score between a complete resume and job description, the framework decomposes the job description into individual requirements and the resume into candidate evidence segments. Each requirement is then evaluated against relevant resume evidence using complementary lexical and semantic matching signals.

CareerFit distinguishes between different types of requirements because they may require different forms of evidence. Explicit or structured requirements, such as programming languages, technologies, degrees, or certifications, depend more strongly on direct terminology and recognized aliases. Experience, responsibility, and behavioral requirements may require greater consideration of lexical and semantic relationships when the same experience is expressed using different wording.

The implemented framework combines keyword and alias matching, TF-IDF similarity, BM25-style lexical relevance, and semantic similarity using requirement-dependent weighting. It also applies an evidence-grounding constraint intended to prevent semantic similarity alone from producing unsupported strong matches for explicit qualifications. For every requirement, the framework identifies the strongest valid resume evidence and classifies the relationship as Matched, Partially Matched, Weakly Supported, or Missing.

The completed framework was evaluated using controlled requirement-evidence cases. In five controlled paraphrase-ranking cases, CareerFit retrieved the expected supporting evidence in **5 out of 5 cases**, compared with **2 out of 5 cases for keyword overlap** and **2 out of 5 cases for TF-IDF cosine similarity**. A separate controlled skill-matching evaluation using 10 manually defined cases produced precision, recall, and F1-score values of **1.000**.

The controlled results suggest that the completed CareerFit framework performed promisingly on the tested paraphrase cases relative to the two reported lexical baselines. In the five controlled cases, CareerFit retrieved the expected supporting evidence more often than keyword overlap and TF-IDF cosine similarity. However, the evaluation datasets are small and manually constructed, so these findings should be interpreted cautiously. They provide preliminary evidence about the behavior of the framework under the tested conditions rather than demonstrating general superiority over other resume-job matching approaches.

The developed CareerFit system provides the practical environment in which the matching framework is implemented and demonstrated. It supports resume upload and editing, job discovery, selected-job comparison, requirement-level reporting, ATS-oriented preparation checks, recommendations, and application tracking. These features constitute the practical contribution of the developed system, while the requirement-aware matching methodology, evidence-grounding mechanism, and controlled comparative evaluation form the main academic contribution.

CareerFit does not attempt to predict whether a candidate will receive an interview or job offer. Instead, it evaluates how strongly the resume provides evidence for the requirements of a selected job and presents the result through an explainable application-readiness workflow.

1. Introduction

Applying for jobs often requires candidates to interpret long job descriptions, identify the most important requirements, and determine whether their resume provides sufficient evidence for each requirement. A resume may contain relevant skills and experience while still failing to express them using the same terminology as the employer.

This creates a matching problem that is more complex than checking whether the same keywords appear in both documents. Some requirements, such as programming languages, technologies, degrees, or certifications, can often be verified using direct terminology or recognized aliases. Other requirements, such as responsibilities, teamwork, communication, or cross-functional collaboration, may be expressed through different wording and therefore require broader lexical or semantic interpretation.

CareerFit investigates this problem through an **explainable requirement-level resume-job matching framework**. Rather than treating a resume and a job description only as two complete documents and producing a single similarity percentage, the proposed approach decomposes the job description into individual requirements and searches the resume for specific evidence supporting each requirement.

The framework combines complementary matching signals, including keyword and alias matching, TF-IDF similarity, BM25-style lexical ranking, and semantic similarity. Their relative importance can vary according to the type of requirement being evaluated. The framework also applies evidence-grounding rules so that a strong match must be supported by identifiable resume evidence rather than by a high similarity score alone.

The research evaluates CareerFit as a comparative framework. Individual matching approaches are treated as baselines and compared with the complete proposed method using controlled, manually labelled requirement-evidence cases. The evaluation considers both whether the correct support classification is produced and whether the correct resume evidence is retrieved.

CareerFit is also implemented as a web-based prototype. The application supports resume input and editing, job discovery, selected-job comparison, explainable reporting, ATS-oriented preparation checks, recommendations, and application tracking. These functions provide the practical environment in which the proposed research framework can be demonstrated and evaluated.

This final report presents the research problem, related work, implemented framework, implementation environment, experimental methodology, controlled experimental results, limitations, and future research directions. The report distinguishes the practical software contribution from the methodological research contribution so that the effectiveness of the proposed matching framework can be evaluated independently from the functionality of the web application.

2. Problem Statement and Objectives

CareerFit addresses the problem of determining whether a candidate's resume provides sufficient and identifiable evidence for the individual requirements of a specific job posting.

Job descriptions commonly contain different types of requirements, including technical skills, tools, education, certifications, experience expectations, responsibilities, and behavioral competencies. Candidates must interpret these requirements and determine whether their resume demonstrates each one clearly. This process can become difficult when applying to several positions or when relevant experience is expressed differently from the terminology used in the job description.

A major limitation of simple resume-job comparison is that relevant evidence may not always contain the same vocabulary as the corresponding job requirement. For example, a requirement such as “collaborate with cross-functional teams” may be supported by a resume statement such as “worked with product, development, and support teams to deliver customer-facing features.” Although the meanings are related, a method based primarily on exact keyword overlap may not recognize the relationship.

At the same time, semantic similarity alone can also produce misleading results. For an explicit qualification such as a certification, degree, or specific technology, generally related experience should not automatically be treated as proof that the requirement has been satisfied.

The problem addressed by CareerFit is therefore not simply how to calculate a similarity score between an entire resume and an entire job description. The project investigates how individual job requirements can be compared with specific resume evidence using complementary lexical and semantic signals while keeping each matching decision understandable and traceable to visible evidence.

CareerFit approaches this problem through requirement-level analysis. The job description is decomposed into individual requirements, the resume is divided into candidate evidence segments, and each requirement is evaluated against the most relevant resume evidence. The resulting decision indicates whether the requirement is Matched, Partially Matched, Weakly Supported, or Missing and identifies the evidence used to support that decision where applicable.

2.1 Problem Statement

The central problem is to determine how strongly a candidate's resume supports the individual requirements of a selected job while avoiding two important limitations of conventional matching approaches.

The first limitation is insufficient recognition of paraphrased evidence. Exact keyword matching and other lexical approaches are transparent and effective when the same terminology appears in both documents, but they may fail when equivalent experience is expressed using different wording.

The second limitation is the possibility of unsupported semantic matches. Semantic similarity can recognize related meanings, but semantic relatedness does not always constitute valid evidence that an explicit qualification is present. For example, cloud-development experience should not by itself demonstrate possession of a required cloud certification.

CareerFit therefore investigates whether different forms of job requirements should be evaluated using different combinations of matching evidence. Explicit and structured requirements may depend more strongly on direct terminology and recognized aliases, while experience, responsibility, and behavioral requirements may benefit from broader lexical and semantic comparison.

The resulting problem can be expressed as follows:

How can a resume-job matching framework identify and classify the evidence supporting each individual job requirement while combining complementary lexical and semantic matching methods and preserving explainability?

2.2 Questions

The project is guided by the following questions:

Q1: How do lexical and semantic matching techniques differ when identifying direct and paraphrased evidence between individual job requirements and resume content?

Q2: Can a requirement-aware hybrid matching framework combine complementary lexical, skill-based, and semantic signals to provide more reliable requirement-level matching than individual matching methods under the same controlled evaluation conditions?

Q3: Can requirement-level matching remain explainable by linking each predicted decision to specific resume evidence and classifying the level of support as Matched, Partially Matched, Weakly Supported, or Missing?

These questions focus the project on the matching methodology and its evaluation rather than on the number of software features implemented in the CareerFit application.

2.3 Objectives

The main objective of CareerFit is to design, implement, and evaluate an explainable requirement-level resume-job matching framework that determines how strongly a candidate's resume provides evidence for the requirements of a selected job.

The project has the following objectives:

1. **Requirement and evidence decomposition:** Decompose job descriptions into individual requirements and resumes into meaningful evidence segments such as skills, experience statements, projects, education, and certifications.
2. **Comparative matching:** Implement complementary matching approaches, including keyword and skill-alias matching, TF-IDF similarity, BM25-style lexical relevance, and semantic similarity, so that their behavior can be compared under consistent conditions.
3. **Requirement-aware matching:** Design and implement a framework in which the relative importance of the matching signals varies according to the type of requirement being evaluated.
4. **Evidence-grounded explanation:** Require requirement-level decisions to be supported by identifiable resume evidence and classify each requirement as Matched, Partially Matched, Weakly Supported, or Missing.

5. **Controlled evaluation:** Evaluate the framework using manually labelled requirement-evidence cases and compare its behavior with individual matching approaches using classification and evidence-retrieval measures where applicable.
6. **Prototype implementation:** Integrate the proposed matching framework into a web-based prototype that allows users to provide a resume and job description, review requirement-level results, identify evidence gaps, and obtain application-readiness information.

The practical application therefore provides the environment in which the matching framework is implemented and demonstrated, while the primary evaluation focuses on the behavior of the requirement-level matching methodology.

The scope of CareerFit is limited to explainable resume-job matching and application-readiness assessment. The project includes resume input and processing, job-description input and limited job discovery, requirement and resume-evidence extraction, requirement-level matching, explainable support classification, ATS-oriented preparation checks, recommendations, and application tracking. CareerFit does not include automatic job-application submission, prediction or guarantee of interview or hiring outcomes, training a new AI model from scratch, large-scale recruitment-platform functionality, unauthorized job-posting scraping, or simulation of a complete commercial ATS.

2.4 Gap and Proposed Contribution

Existing person-job fit approaches have used lexical similarity, semantic representations, neural models, ranking methods, feature fusion, and explainable recommendation techniques. Therefore, CareerFit does not claim that keyword matching, TF-IDF, BM25, semantic similarity, hybrid matching, or explainability are individually new.

The gap addressed by CareerFit is more specific. Many resume-job matching approaches focus primarily on an overall resume-job fit score, candidate ranking, or document-level similarity. CareerFit instead investigates the problem from the perspective of an individual job seeker asking:

For each requirement in this specific job posting, what evidence does my resume provide, how strong is that evidence, and why was that decision made?

CareerFit addresses this problem through a requirement-aware and evidence-grounded matching framework.

The proposed contribution has three main elements.

First, **requirement-level analysis** changes the unit of comparison from the complete resume-job pair to individual requirement-evidence relationships. This allows important missing requirements to remain visible even when the resume is otherwise strongly aligned with the position.

Second, **requirement-aware matching** allows the importance of lexical, skill-based, and semantic evidence to vary according to the nature of the requirement. Explicit qualifications rely more strongly on direct terminology and recognized aliases, while responsibilities and behavioral requirements can place greater emphasis on lexical-semantic evidence.

Third, **evidence grounding** requires the framework to identify specific resume evidence supporting a positive requirement-level decision. For explicit qualifications such as technologies, degrees,

licenses, or certifications, semantic similarity alone is not treated as sufficient evidence for the strongest support classification.

The resulting process can be summarized as:

Job Requirement → Resume Evidence → Matching Signals → Requirement-Aware Scoring → Evidence Grounding → Support Classification → Explanation

The contribution of CareerFit is therefore not the individual matching algorithms used inside the system, but the organization and evaluation of these complementary methods within a transparent requirement-level decision framework.

The framework is evaluated using controlled, manually labelled requirement-evidence cases. The evaluation considers both whether the expected requirement relationship is identified and whether the correct supporting resume evidence is retrieved. This makes it possible to study matching effectiveness and explainability together rather than evaluating the system only through a final similarity percentage.

3. Technology and Method Selection

The technologies and matching methods used in CareerFit were selected to support both the implementation of the web-based prototype and the requirement-level resume-job matching process.

The software stack was chosen to provide a modular client-server architecture in which resume processing, job handling, matching, reporting, and data persistence could be implemented as separate components. The matching methods were selected because they provide complementary forms of lexical and semantic evidence and can be evaluated independently under the same controlled conditions.

Area	Technology / Method	Purpose in CareerFit
Frontend	React with Vite [12]	Provides the user interface for resume input, job discovery, matching results, reports, and application tracking.
Styling	Tailwind CSS / standard CSS	Supports a consistent and responsive user interface.
Backend	Django and Django REST Framework [10], [11]	Handles authentication, resume processing, job retrieval, matching requests, report generation, and persistence.
Database	SQLite for local development; PostgreSQL-ready architecture	Stores user, resume, job, report, and application-tracking data.
Resume Parsing	pypdf, python-docx, and TXT parsing	Extracts text from supported PDF, DOCX, and TXT resume formats [19], [20].
Job Data Sources	Adzuna, Remotive, Arbeitnow, Jooble, and fallback data where necessary	Supports the practical job-discovery workflow where provider data is available [6]–[8], [18]. Manual job-description input remains available as a fallback.
Keyword / Alias	Python-based normalization,	Provides transparent direct evidence for explicit

Matching	skill dictionaries, and aliases	skills and terminology.
TF-IDF Similarity	TF-IDF vectorization and cosine similarity	Provides weighted lexical comparison between job requirements and resume evidence [3].
BM25-Style Ranking	Information-retrieval ranking logic	Provides an additional lexical relevance signal for ranking candidate resume evidence.
Requirement Typing	Rule-based requirement classification	Identifies requirement categories used to determine the relative importance of matching signals.
Semantic Matching	Controlled semantic concepts and optional embedding-based similarity	Supports recognition of related or paraphrased evidence when vocabulary overlap is limited.
Requirement-Aware Scoring	CareerFit scoring logic	Combines the matching signals according to requirement type and applies evidence-grounding rules.
Evaluation	Python evaluation scripts and manually labelled cases	Supports controlled comparison of matching methods and evidence-retrieval evaluation.
Software Testing	Django tests, frontend unit tests, build checks, and end-to-end tests	Verifies application functionality independently from matching evaluation.
Optional AI Coaching	OpenAI or Ollama, when configured	Provides optional user-facing coaching after the deterministic analysis and is not required for the core matching process.

The core CareerFit matching process is designed to remain transparent and reproducible. Keyword and alias matching, TF-IDF, BM25-style lexical relevance, requirement typing, requirement-aware weighting, and evidence-grounding rules can operate independently of an external generative AI service.

Semantic similarity is used as an additional evidence signal when broader meaning is required, particularly for responsibilities and behavioral requirements. Its influence is controlled according to requirement type so that semantic relatedness does not replace direct evidence for explicit qualifications.

Optional AI-assisted coaching is kept separate from the matching and scoring process. This separation allows the main CareerFit decisions to remain traceable to identifiable matching signals and resume evidence.

4. Implemented Solution

CareerFit was implemented as a web-based resume-job matching system that helps users evaluate how well their resume fits a specific job posting or target role. The system guides users through a practical workflow: create or update a profile, upload or paste a resume, review extracted resume text, search for jobs or provide a job description, select a posting, generate a preview, produce a detailed readiness report, and save the role in an application tracker.

The system supports multiple job-input paths. A user may paste a job description manually, import a public URL where accessible, or search for jobs inside the application by title, location, country, source, workplace type, skills, experience level, employment type, and salary. Once a job is selected, CareerFit loads the job description and compares it directly with the current resume. This connected workflow is a major practical contribution of the implemented prototype and provides the application environment in which the proposed matching framework is demonstrated.

4.1 Main Features

Feature	Implemented Functionality
Resume input	User can upload PDF, DOCX, TXT, or paste resume text manually.
Editable resume preview	User can review and correct extracted resume text before analysis.
Job input	User can paste a job description, import a public URL, or search by job title.
Requirement extraction	System identifies skills, tools, responsibilities, education, experience, soft skills, and certifications.
Matching framework	System compares resume evidence with job requirements using multiple methods.
Explainable report	System shows matched, partially matched, weak, and missing requirements.
Readiness score	System combines content alignment with ATS-oriented preparation checks.
Application tracker	User can save roles and track application progress.
Optional AI coaching	User may request additional coaching after deterministic analysis.

4.2 Research and Practical Contribution

CareerFit has two distinct contributions: a **research contribution** related to resume-job matching methodology and a **practical contribution** related to application-readiness support. Separating these contributions is important because the web application itself is not considered the primary research contribution. Instead, the application provides an experimental environment in which the proposed matching framework can be implemented, evaluated, and demonstrated.

4.2.1 Research Contribution

The main research contribution of CareerFit is the proposed CareerFit Requirement-Aware Explainable Matching Framework, which evaluates resume-job compatibility at the level of individual job requirements rather than relying only on whole-document similarity.

The framework is motivated by the observation that job requirements are heterogeneous. A technical skill requirement, an experience requirement, an educational qualification, and a soft-skill responsibility do not necessarily require the same type of matching evidence.

For example, the requirement:

“Experience with Python”

can normally be supported by direct evidence such as “Python,” recognized skill aliases, or statements describing Python-based development.

In contrast, the requirement:

“Ability to collaborate with cross-functional teams”

may be supported by a resume statement such as:

“Worked with product, development, and support teams to deliver customer-facing features.”

The second example contains limited exact vocabulary overlap even though the meaning is related. Consequently, applying the same lexical similarity rule to both requirements may not provide the most appropriate evidence.

CareerFit therefore defines the matching task as a sequence of requirement-level decisions. For each extracted job requirement, the framework:

1. identifies the requirement and its relevant type or characteristics;
2. extracts candidate evidence segments from the resume;
3. evaluates the requirement-evidence relationship using complementary matching signals;
4. identifies the strongest supporting resume evidence;
5. calculates a requirement-level support score;
6. classifies the relationship as **Matched**, **Partially Matched**, **Weakly Supported**, or **Missing**; and
7. returns both the classification and the evidence used to support the decision.

The framework can use multiple matching signals, including exact keyword and alias matching, lexical similarity such as TF-IDF and BM25-style ranking, and semantic similarity. These techniques are not presented as original algorithms individually. The research contribution concerns how these complementary signals are organized within a requirement-level, evidence-grounded decision framework.

A central design principle is evidence grounding. A high similarity score alone should not be sufficient to classify a requirement as strongly supported. The system should also identify a resume segment that provides interpretable evidence for the decision. This allows the user to examine why the system produced a particular classification.

A second contribution is the separation of job-match assessment from application-readiness assessment. The job-match component evaluates whether resume content provides evidence for job requirements. ATS-oriented preparation checks evaluate a different question: whether the resume is structurally prepared for submission. These dimensions are therefore presented separately before being combined into an overall readiness measure.

4.2.2 Comparative Research Contribution

CareerFit is designed as a comparative framework rather than assuming that one matching technique is always superior.

The CareerFit framework is evaluated using the same controlled resume-requirement cases as the reported baseline approaches so that method behavior can be compared under consistent experimental conditions. The main baselines are:

- keyword and skill-alias matching;
- TF-IDF similarity;
- BM25-style lexical ranking; and
- semantic similarity.

The broader validation methodology covers direct vocabulary matches, partial evidence, missing requirements, paraphrased evidence, and semantic false-positive cases. The completed quantitative results reported in Section 13 are limited to the controlled cases actually executed and recorded.

The purpose of the evaluation is to determine when each technique succeeds or fails, and whether the complete CareerFit framework provides a better balance between matching effectiveness and explainability than the individual baseline methods.

The proposed framework should therefore not be evaluated only by the final match percentage. Evaluation should consider whether the predicted requirement classification is correct and whether the system retrieves the correct resume evidence supporting that classification.

This distinction is important because two systems could produce similar overall match scores while providing very different explanations at the requirement level.

4.2.3 Practical Contribution

The practical contribution of CareerFit is a connected application-readiness workflow built around the research framework.

The prototype allows a user to provide a resume, review extracted resume content, search for or provide a job description, select a target position, perform requirement-level resume-job analysis, review the supporting evidence, identify gaps, and save the selected opportunity for tracking.

The practical interface is therefore not intended to represent the primary academic novelty of the project. Its role is to make the matching framework usable and observable in a realistic job-application scenario.

CareerFit also provides recommendations connected to identified evidence gaps. If a requirement is missing or weakly supported, the recommendation can identify the relevant problem rather than presenting only a generic resume-improvement suggestion.

4.2.4 Contribution Summary

The contribution of CareerFit can therefore be summarized at three levels:

Methodological contribution: A requirement-level hybrid matching framework that evaluates different forms of evidence rather than treating the complete resume and job description only as two documents.

Explainability contribution: Each requirement-level decision is grounded in visible resume evidence and represented using an interpretable support category.

Evaluation contribution: The framework provides a controlled environment for comparing lexical and semantic matching approaches on direct, partial, missing, and paraphrased resume-job evidence.

These contributions define the research value of CareerFit independently from the amount of software development required to implement the web application.

The implemented prototype provides the software environment required to execute and demonstrate the CareerFit matching process. The application integrates resume processing, job input and discovery, requirement extraction, requirement-level matching, explainable reporting, and application tracking within a connected workflow. However, successful implementation alone does not demonstrate the effectiveness of the matching approach. The behavior of the matching methods and the strength of the resulting conclusions are evaluated separately through the experimental methodology and controlled results presented later in the report.

5. Limitations

CareerFit was developed as an academic prototype and should be interpreted within both implementation and evaluation limitations.

From an implementation perspective, the availability and completeness of external job-provider data may vary, which can affect the job-discovery functionality. To reduce this dependency, CareerFit also allows users to provide a job description manually or import a publicly accessible posting where supported.

Resume parsing may also be imperfect, particularly for PDF documents with complex formatting, multiple columns, tables, or unusual layouts. For this reason, users can review and edit the extracted resume text before performing the analysis.

The main evaluation limitation is the small size of the controlled dataset. The reported experiments use manually constructed requirement-evidence cases designed to examine specific behaviors such as direct matches, paraphrased evidence, missing requirements, and potential semantic false positives. These cases are useful for controlled comparison, but they do not represent the full diversity of resumes, occupations, industries, experience levels, and job-description styles found in real recruitment settings. The reported results should therefore be interpreted as evidence under the tested conditions rather than as a general benchmark for resume-job matching.

The expected requirement classifications and supporting resume evidence are also defined manually. This introduces a degree of subjectivity, particularly when distinguishing between Matched, Partially Matched, and Weakly Supported requirements. Different evaluators may not always agree on the strength of evidence for broad experience or behavioral requirements. A stronger evaluation would require multiple independent annotators and measurement of inter-annotator agreement.

The requirement-aware scoring model uses predefined signal weights and fixed classification thresholds. These values are design parameters rather than parameters learned from a large labelled dataset. They were fixed before the reported controlled comparison to reduce the risk of adjusting them to favor the final results; however, broader sensitivity analysis is required to determine whether the observed behavior remains stable when reasonable variations in the weights and thresholds are introduced.

The framework also depends on assigning each requirement to a broad category such as Explicit / Structured, Experience / Responsibility, Behavioral / Soft Skill, or General / Ambiguous. Some requirements may contain characteristics of more than one category, and incorrect requirement typing may affect the relative influence assigned to the matching signals.

Semantic matching introduces an additional limitation. It can improve the recognition of paraphrased evidence when vocabulary overlap is limited, but semantic relatedness can also produce false-positive relationships. For example, general cloud-computing experience may be semantically related to an AWS certification without demonstrating that the candidate actually holds the certification. CareerFit uses evidence-grounding rules to reduce this risk, particularly for explicit skills, degrees, licenses, and certifications, but broader evaluation is required to determine how reliably these controls operate across different requirement types.

The current external benchmarking is also limited. Although the evaluation methodology defines reproducible baselines such as keyword/alias matching, TF-IDF, BM25-style ranking, and semantic similarity, the completed quantitative comparison does not report numerical results for every defined baseline. In addition, complete published systems such as PJFNN, IPJF, ConFit, and JobMatchAI are not reproduced under identical experimental conditions, and commercial tools such as Jobscan and SkillSyncer are not directly benchmarked because their internal algorithms and evaluation conditions are not fully available.

Finally, CareerFit evaluates resume-job alignment and application readiness. Its scores should not be interpreted as predictions or guarantees of interview selection, job offers, or other hiring outcomes.

6. Related Work and Research Positioning

Resume-job matching, also referred to as Person-Job Fit (PJF), has been studied using information-retrieval methods, machine-learning models, neural representation learning, semantic matching, ranking approaches, and more recently hybrid retrieval architectures. CareerFit builds on this research rather than treating resume-job matching as a new problem.

The purpose of this section is therefore to identify how existing approaches formulate the matching problem and to position the proposed CareerFit framework relative to them.

6.1 Existing Person-Job Fit Approaches

Zhu et al. proposed the Person-Job Fit Neural Network (PJFNN), an end-to-end CNN-based model that learns joint representations of job requirements and candidate qualifications from historical job applications [26]. The model can estimate overall person-job fit while also identifying which requirement items are satisfied by a candidate. This work demonstrates that requirement-level relationships can be learned through neural representations. However, the approach is primarily a learned prediction model and depends on historical application data.

Le et al. proposed the Interpretable Person-Job Fit (IPJF) model [27]. Their work explicitly addresses interpretability and models the perspectives of both employers and job seekers using multi-task optimization and deep interactive representation learning. The system formulates matching as a learning-to-rank problem and demonstrates that interpretability is already an established research concern in person-job fitting. Therefore, CareerFit does not claim that explainability itself is new.

Menacer et al. proposed an interpretable hybrid person-job fitting approach that combines several human-readable and semantic features [28]. Their approach includes count-based features, TF-IDF similarity, semantic representation, explicit skill matching, experience-related features, and other resume characteristics. The features are combined through weighted scoring or machine-learning models for candidate classification and ranking. This work is particularly relevant to CareerFit because it demonstrates that combining lexical, semantic, and skill-related features is also not sufficient by itself to establish novelty.

More recent research has focused on learned semantic representations and large-scale ranking. Yu et al. introduced ConFit, which uses data augmentation and contrastive learning for resume-job matching [29]. ConFit generates additional resume-job training examples through paraphrasing and uses contrastive learning to address sparse interaction data. The method was evaluated on real-world

resume-job datasets against retrieval and embedding baselines. Its main research objective is effective resume/job retrieval and ranking rather than requirement-level application-readiness explanation.

Recent work also demonstrates increasingly sophisticated hybrid architectures. Vyas et al. introduced JobMatchAI, which combines BM25 retrieval, Transformer-based semantic representations, a skill knowledge graph, and interpretable reranking [30]. The system evaluates multiple factors such as skills, experience, location, and other preferences and provides factor-wise explanations. This is important for the positioning of CareerFit because it shows that hybrid lexical-semantic matching and explainability already exist in current job-matching research.

6.2 Comparative Positioning of CareerFit

The previous studies demonstrate that neither semantic matching, hybrid feature fusion, requirement awareness, nor explainability can individually be claimed as a completely new idea. CareerFit is therefore positioned through the specific research task it investigates: **requirement-level, evidence-grounded application-readiness assessment using a lightweight and transparent hybrid framework**.

The following table summarizes the methodological differences between representative existing approaches and the proposed CareerFit framework.

Approach	Main Matching Approach	Primary Output	Explainability / Granularity	Main Distinction from CareerFit
PJFNN — Zhu et al. [26]	CNN-based joint representation learning from historical applications	Person-job fit prediction	Can identify satisfied requirement items through learned representations	CareerFit does not train a person-job prediction model from historical applications; it retrieves explicit resume evidence and produces interpretable support categories.
IPJF — Le et al. [27]	Deep interactive representation learning and multi-task learning-to-rank	Candidate/job ranking	Provides interpretable recommendation reasoning from employer and job-seeker perspectives	CareerFit focuses on job-seeker application readiness and explicit requirement-to-resume evidence rather than reciprocal recommendation ranking.
Menacer et al. [28]	Fusion of count, TF-IDF, semantic, skill and resume-specific features	Candidate classification and ranking	Uses human-readable features to improve interpretability	CareerFit focuses on individual requirement-evidence decisions and evaluates whether the returned resume segment supports each requirement.
ConFit — Yu et al. [29]	Data augmentation and contrastive	Resume/job retrieval and ranking	Primarily optimized for matching and ranking	CareerFit emphasizes transparent requirement-level decisions and does

	representation learning		performance	not require training a task-specific contrastive model.
JobMatchAI — Vyas et al. [30]	BM25, semantic embeddings, skill knowledge graph and interpretable reranking	Job/candidate retrieval and ranking	Factor-wise explanations across multiple matching dimensions	CareerFit investigates evidence-grounded support for individual job requirements and classifies each as Matched, Partial, Weak, or Missing.
Proposed CareerFit Framework	Requirement-aware combination of lexical, skill-based and semantic evidence signals	Requirement-level support assessment and application readiness	Explicit job requirement → resume evidence → support classification	Designed to study whether requirement-aware evidence grounding provides an effective and transparent application-readiness assessment without relying on a single global similarity or ranking score.

6.3 Gap Addressed by CareerFit

The literature shows that considerable research already exists on person-job fit prediction, candidate ranking, semantic matching, feature fusion, and interpretable recommendation. Consequently, the research gap addressed by CareerFit must be defined more narrowly than simply combining multiple matching algorithms.

CareerFit investigates the problem from the perspective of a job seeker asking a different question:

“For each requirement in this specific job posting, what evidence does my resume provide, how strong is that evidence, and why was that decision made?”

This changes the unit of analysis from an overall resume-job pair or candidate-ranking score to an individual **requirement-evidence relationship**.

The proposed framework therefore models the matching process as:

Job Requirement → Candidate Resume Evidence → Matching Signals → Support Score → Support Classification → Explanation

A key research design decision is that matching evidence is evaluated according to the nature of the requirement. Exact technical requirements can be strongly supported by direct terms and recognized aliases, while responsibilities and soft-skill requirements may require greater consideration of lexical-semantic relationships.

CareerFit also introduces an evidence-grounding constraint into the application-readiness workflow. A high global similarity between a resume and job description is not treated as sufficient evidence

that a particular requirement is satisfied. The framework attempts to retrieve a specific resume segment that supports the predicted requirement relationship.

The research evaluation therefore defines two related outcomes:

1. **Requirement classification correctness:** whether the framework correctly identifies a requirement as Matched, Partially Matched, Weakly Supported, or Missing.
2. **Evidence retrieval correctness:** whether the resume segment selected by the framework corresponds to the evidence expected to support the requirement.

The experimental contribution of CareerFit is the comparison of this complete requirement-aware framework with its individual matching components under the same manually labelled cases. This allows the project to investigate not simply which algorithm produces the highest similarity value, but which approach provides the most useful combination of matching effectiveness and interpretable evidence.

6.4 Relationship to Commercial Resume Tools

Commercial platforms such as Jobscan, SkillSyncer, Resume Worded, and Teal remain relevant as practical examples of resume optimization, keyword analysis, ATS-oriented checking, and application-management workflows [14]–[17].

However, these systems are used in this report primarily to provide **practical product context**, not as the main scientific baselines for the research evaluation. The experimental comparison in CareerFit focuses on reproducible matching methods—keyword/alias matching, TF-IDF, BM25, semantic similarity, and the proposed CareerFit framework—because these methods can be implemented under the same conditions and evaluated using the same labelled cases.

This distinction separates the commercial comparison from the academic comparison and provides clearer criteria for evaluating CareerFit as a research project.

7. Target Users and Use Cases

CareerFit is designed for job seekers who want to prepare stronger applications for specific roles. The primary user has a resume, wants to compare it with a job posting, and needs clear feedback before applying.

7.1 User Persona

Attribute	Description
Name	Sara
Profile	Early-career professional preparing targeted job applications
Goal	Improve her resume before applying to selected roles
Current behavior	Reads job descriptions manually and keeps several resume versions
Main difficulty	Long job descriptions make it hard to identify the most important gaps
Design need	Clear workflow, plain-language feedback, visible evidence, and editable results

7.2 Main Use Cases and User Stories

Use Case / Story	Description
Upload resume	User uploads a PDF, DOCX, or TXT resume and reviews the extracted text.
Edit resume text	User corrects extracted text before analysis.

Search or import jobs	User searches by title/location, imports a URL, or pastes a job description.
Quick comparison	User receives a preliminary match summary.
Generate report	User receives scores, evidence, ATS checks, and recommendations.
Track application	User saves a selected role and records status, notes, dates, and tasks.
Optional coaching	User may request AI-assisted suggestions after deterministic analysis.

8. Functional and Non-Functional Requirements

The requirements are based on the approved proposal and focus on the behavior, usability, explainability, privacy, reliability, and evaluation needs of CareerFit.

8.1 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow users to create an account, log in, log out, and manage a profile.
FR-02	The system shall allow users to upload a resume file or paste resume text manually.
FR-03	The system shall support PDF, DOCX, TXT, and pasted text input.
FR-04	The system shall allow users to review and edit extracted resume text before analysis.
FR-05	The system shall allow users to paste a complete job description manually.
FR-06	The system shall allow users to import a public job posting URL where accessible.
FR-07	The system shall allow users to search a limited number of postings by job title and location.
FR-08	The system shall extract important job requirements from job descriptions.
FR-09	The system shall extract candidate evidence from resume content.
FR-10	The system shall compare resume evidence with job requirements.
FR-11	The system shall classify requirements as matched, partially matched, weakly supported, or missing.
FR-12	The system shall calculate a match score and an application-readiness score.
FR-13	The system shall display ATS-oriented resume checks separately from content alignment.
FR-14	The system shall provide practical recommendations for improving the resume.
FR-15	The system shall summarize recurring skills from retrieved job postings.
FR-16	The system shall allow users to save selected jobs in an application tracker.
FR-17	The system shall support optional AI-assisted coaching after deterministic analysis.
FR-18	The system shall support controlled comparison of keyword/alias matching, TF-IDF similarity, BM25-style ranking, semantic similarity, and the proposed CareerFit framework using the same labelled requirement-evidence cases.

8.2 Non-Functional Requirements

ID	Non-Functional Requirement	Design Response
NFR-01	Explainability	Scores should be supported by visible requirement-level evidence.
NFR-02	Usability	The interface should guide users step by step from resume upload to report generation.
NFR-03	Privacy	Resume data and private user records should be connected to the authenticated user.
NFR-04	AI transparency	Optional AI coaching should require explicit user action and should not replace the deterministic score.
NFR-05	Maintainability	Frontend, backend, parsing, matching, job search, and reporting responsibilities should be separated.
NFR-06	Reliability	Fallback options should exist when job-provider data is incomplete or unavailable.
NFR-07	Testability	The prototype should support backend tests, frontend tests, and evaluation scripts.

8.3 Acceptance Criteria

ID	Acceptance Criterion
AC-01	A valid PDF, DOCX, TXT, or pasted resume produces editable resume text.
AC-02	An unsupported or invalid file produces a clear error message.
AC-03	The user can correct extracted resume text before running a comparison.
AC-04	The user can search for jobs by title and location or paste a job description.

AC-05	The detailed report separates matched, partial, weak, and missing requirements.
AC-06	Each matched or partially matched requirement includes visible resume evidence where available.
AC-07	The report shows both match score and application-readiness score.
AC-08	ATS checks are shown separately from content-alignment results.
AC-09	Recommendations are linked to missing or weak areas.
AC-10	The research evaluation shall compare the proposed CareerFit framework with keyword/alias matching, TF-IDF, BM25-style ranking, and semantic similarity using the same labelled requirement-evidence cases.
AC-11	For each supported requirement, the framework shall return the resume evidence segment used to justify the predicted support classification.
AC-12	The evaluation shall measure both requirement-classification correctness and evidence-retrieval correctness.

8.4 Requirement and Research Implementation Status

The functional application requirements and the revised CareerFit research framework have now been integrated into the working prototype. The core application supports resume processing, job discovery, requirement extraction, requirement-aware matching, evidence grounding, explainable classification, reporting, and application tracking. Research evaluation is reported separately from implementation status because successful implementation does not by itself demonstrate matching effectiveness.

Requirement Area	Status	Notes
Authentication and Profile	Implemented	Registration, login, account management, and profile functionality are available.
Resume Input and Parsing	Implemented	PDF, DOCX, TXT, and pasted resume text are supported, with editable extracted content.
Job Discovery and Input	Implemented with provider limitations	Job search, manual job-description input, fallback data, and public URL import are supported where accessible.
Requirement Extraction	Implemented	Rule-based requirement extraction is integrated into the requirement-level matching framework.
Resume Evidence Segmentation	Implemented	Resume content is segmented into evidence areas such as experience, skills, projects, education, and certifications for requirement-level comparison.
Keyword / Alias Matching	Implemented	Provides the direct lexical signal and an independent baseline for explicit terminology and recognized aliases.
TF-IDF Similarity	Implemented	Provides the TF-IDF lexical-similarity signal and experimental baseline.
BM25-Style Ranking	Implemented	BM25-style lexical relevance is integrated as one of the four matching signals and can also be evaluated independently as a baseline.
Semantic Matching	Implemented core semantic matching	Controlled semantic matching is integrated into the framework. Embedding-based similarity remains an optional extension and is not required for the core method.
Requirement-Type Identification	Implemented	Requirements are categorized as Explicit / Structured, Experience / Responsibility, Behavioral / Soft Skill, or General / Ambiguous before signal weighting.
Requirement-	Implemented	Keyword/Alias, TF-IDF, BM25, and semantic signals

Aware Weighting		are combined using the requirement-dependent weights defined in Section 11.5.1.
Evidence-Grounding Constraint	Implemented	Explicit qualifications require appropriate direct evidence, preventing semantic similarity alone from producing the highest support classification.
Requirement-Level Classification	Implemented	The revised scoring model produces Matched, Partially Matched, Weakly Supported, and Missing classifications using the thresholds defined in Section 11.5.4.
Explainable Reporting	Implemented	Requirement-level classifications and supporting resume evidence are available to the reporting layer for explainable presentation.
Application Tracker	Implemented	Selected jobs, status, notes, dates, and application information can be stored and updated.
Research Evaluation	Controlled evaluation performed; broader validation remains limited	The completed framework has been evaluated on the reported controlled cases. Larger-scale validation, expanded baseline analysis, sensitivity analysis, and ablation remain limitations or future work where not yet performed.

This distinction prevents completed software functionality from being confused with research effectiveness. The CareerFit application and requirement-aware matching framework are implemented, while the strength of the research conclusions is limited to the controlled evaluation reported in Section 13.

9. System Architecture and User Interface

CareerFit uses a client-server architecture that separates the user-facing application from the backend services responsible for resume processing, job handling, matching, scoring, reporting, and data persistence.

The React/Vite frontend provides the user interface through which users manage their profile, provide and edit resume content, search for or enter job descriptions, review matching results, and track applications.

The Django REST Framework backend manages authentication, resume parsing, job retrieval, database operations, and the matching workflow.

Within this architecture, the CareerFit Requirement-Aware Explainable Matching Framework operates as a dedicated backend component. It receives processed job requirements and resume evidence, calculates the matching signals, applies requirement-aware scoring and evidence-grounding rules, and returns requirement-level results to the reporting layer.

Separating the matching component from the user-interface logic allows the matching methodology to be evaluated independently from the other application functions.

SQLite is used for local prototype development, while the architecture remains compatible with PostgreSQL for future deployment. External job providers and optional semantic or coaching services

are accessed through separate backend service modules so that the core matching workflow can operate independently of these external services.

9.1 Client-Server Architecture

The CareerFit architecture is divided into three main layers: the frontend interface, backend application services, and data-storage layer.

The frontend communicates with the backend through REST-based requests. User actions such as uploading a resume, selecting a job, generating an analysis, or updating application information are processed by the corresponding backend modules.

The backend separates general application functionality from the matching logic. Resume-processing services prepare candidate evidence, job-processing services prepare job requirements, and the matching component performs requirement-level comparison. The resulting classifications, evidence mappings, scores, and recommendations are then returned to the reporting interface.

This modular organization supports maintainability and also allows the matching framework to be tested independently from the rest of the web application.

9.2 Backend Modules

Module	Responsibility
Accounts	Registration, login, logout, profile management, and authentication.
Resumes	Resume upload, parsing, normalization, editing, and saved resume versions.
Jobs	Job search, URL import, selected roles, role information, and application-tracker records.
Matching	Requirement extraction and typing, resume-evidence segmentation, keyword/alias matching, TF-IDF similarity, BM25-style ranking, semantic evidence analysis, requirement-aware scoring, evidence grounding, support classification, baseline evaluation, and report-data generation.
Coaching	Optional AI-assisted recommendations after user consent.

9.3 Security and Privacy Considerations

Resume content and user application information may contain sensitive personal data. CareerFit therefore associates private records with authenticated users and separates authentication and data-management responsibilities within the backend.

Application secrets and external-service credentials should be stored through environment variables rather than directly in the source code.

Optional AI-assisted coaching requires explicit user action before resume or job information is sent to an external model. The core CareerFit matching and scoring process does not depend on an external generative AI service.

9.4 Implemented User Interfaces

The implemented interfaces provide the practical workflow through which users interact with CareerFit. Rather than treating each screen as a separate system component, the interfaces are

presented together because they represent different stages of the same application-readiness workflow.

The main implemented interfaces are:

- **Candidate Profile:** allows the user to manage profile-related information.
- **Resume Workspace:** supports resume upload, extracted-text review, and editing.
- **Job Discovery:** allows users to search for jobs and select a target position.
- **Resume-Job Comparison Preview:** provides an initial comparison for the selected job.
- **Application Readiness Report:** displays requirement-level classifications, supporting resume evidence, scores, ATS-oriented checks, and recommendations.
- **Application Tracker:** allows users to save selected jobs and manage application status and related information.

9.4.1 Home Page Interface

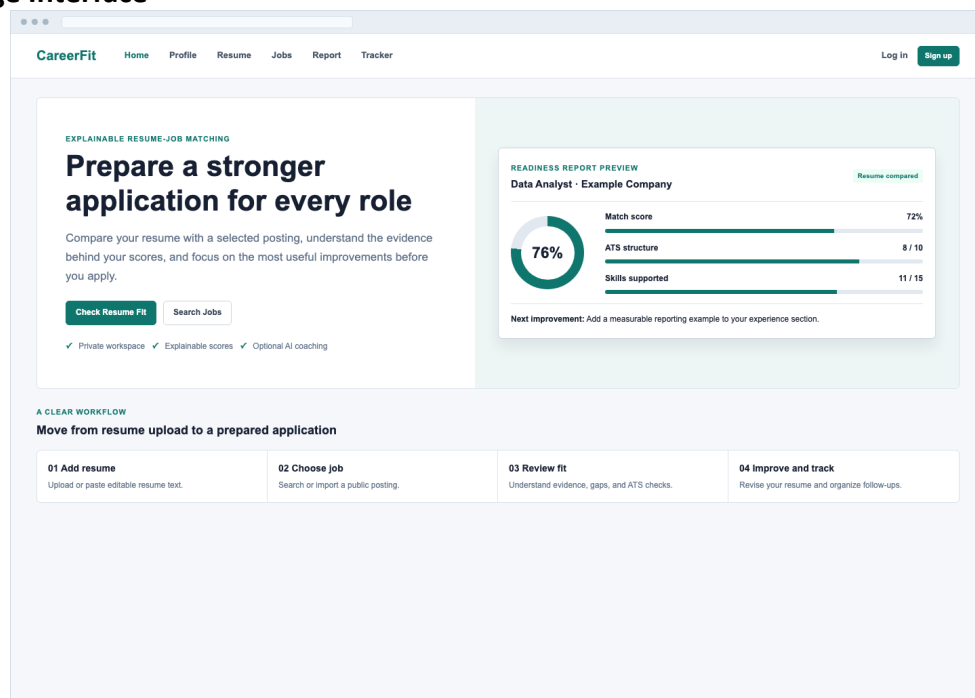


Figure 1. CareerFit home page and workflow entry point.

9.4.2 Candidate Profile Interface

The screenshot shows the 'Candidate Profile' page in the CareerFit application. The top navigation bar includes 'Home', 'Profile', 'Resume', 'Jobs', 'Report', and 'Tracker'. The main heading is 'Candidate Profile' with a subtext 'Add the information that will personalize your job search and readiness report.' A 'Profile completion' progress bar shows 75%. The form is divided into three main sections: 'Contact details' (Name, Email, Phone, Location), 'Career preferences' (Target role, Experience level, Workplace preference), and 'Professional summary' (a text area). A 'Profile status' sidebar on the right lists completed items (Name and contact details, Target role, Location) and a pending item (Add a stronger summary), with a 'Save Profile' button at the bottom.

CareerFit Home Profile Resume Jobs Report Tracker Sample workspace

CANDIDATE PROFILE

Candidate Profile

Add the information that will personalize your job search and readiness report.

ID Contact details
Basic information for your private workspace.

Name: Sample Candidate Email: candidate@example.com
Phone: +1 555 010 2026 Location: Montreal, QC

JOB Career preferences
Choose a role and working style to guide job discovery.

Target role: Data Analyst Experience level: Entry level Workplace preference: Hybrid preferred

TX Professional summary
Describe the type of work you want to pursue.

Analytical candidate interested in reporting, data visualization, and clear communication of business insights.

Profile status

- ✓ Name and contact details
- ✓ Target role
- ✓ Location
- ! Add a stronger summary

Save Profile

Figure 2. Candidate profile interface.

9.4.3 Resume Workspace Interface

The screenshot shows the 'Resume Workspace' page in the CareerFit application. The top navigation bar includes 'Home', 'Profile', 'Resume', 'Jobs', 'Report', and 'Tracker'. The main heading is 'Resume Workspace' with a subtext 'Upload or paste a resume, then review the extracted text before comparing it with jobs.' The interface is divided into three main sections: 'Add your resume' (with a file upload area and 'Browse files'/'Paste resume text' buttons), 'Editable resume text preview' (showing extracted text for contact details, professional summary, skills, and experience), and 'ATS preparation checklist' (listing items like Contact information, Summary section, Skills section, Add measurable results, and Review date formatting). A 'Saved resume versions' sidebar on the right allows saving a tailored version for another role. The bottom of the preview section includes a word count (59 words / 428 characters) and 'Clear Resume'/'Continue to Jobs' buttons.

CareerFit Home Profile Resume Jobs Report Tracker Sample workspace

RESUME PREPARATION

Resume Workspace

Upload or paste a resume, then review the extracted text before comparing it with jobs.

UP Add your resume
Accepted formats: PDF, DOCX, and TXT.

Drop your resume file here
or choose a file from your computer

Browse files **Paste resume text**

Editable resume text preview
Correct extraction issues before continuing.

Resume loaded

SAMPLE CANDIDATE
Montreal, QC - candidate@example.com

PROFESSIONAL SUMMARY
Analytical candidate with experience preparing reports and visual dashboards.

SKILLS
SQL · Python · Excel · Tableau

EXPERIENCE
Reporting Project - Example Organization
- Prepared a dashboard and presented findings to project stakeholders.

59 words · 428 characters **Clear Resume** **Continue to Jobs**

ATS preparation checklist

- ✓ Contact information
- ✓ Summary section
- ✓ Skills section
- ! Add measurable results
- ! Review date formatting

ATS structure 7 / 9

Saved resume versions
Reuse a tailored version for another role.

Version name **Save Version**

General Resume
Updated recently

Figure 3. Resume workspace with upload and editable extracted text.

9.4.4 Job Discovery Interface

CareerFit
Application readiness

Home Dashboard Profile Resume **Jobs** Report Tracker

Figure Demo Student Sign out

Filters and alerts

Website: Sample demo Workplace: Any workplace Skills: Python SQL Exclude keywords: senior, unpaid Experience: Any level Employment: Any type Salary min: 50000 Salary max: 90000

Save search alert

ACTIVE FILTERS Website: Sample demo Clear filters

Live job providers are unavailable. Showing local sample postings.

Role insights from 2 retrieved postings

problem solving - 2/2 python - 2/2 sql - 2/2 api - 1/2 aws - 1/2 communication - 1/2 dashboards - 1/2 django - 1/2

RELATED ROLES Junior Data Analyst Junior Software Engineer

Search results 2 results - page 1

Junior Data Analyst Sample Posted May 30

Northstar Analytics
Remote, United States
\$55,000 - \$70,000

Remote Full-Time Entry

We are looking for a Junior Data Analyst who can collect, clean, and analyze business data. Required skills include Python, SQL, Excel, Tableau or Power BI, communication, and problem solving. The candidate should create dashboards, explain findings to stakeholders, and work with cross-functional teams.

Compare with resume

Junior Software Engineer Sample Posted May 20

Civic Cloud
New York, United States
\$68,000 - \$82,000

On-Site Full-Time Entry

Build web services and internal tools using Python, Django, REST APIs, SQL, Docker, Git, and AWS. Strong problem solving.

Resume comparison
Select a job to check your fit.

Job text used for scoring
CareerFit scores the job description text only. Title, company, location, and source links are kept separately so metadata does not affect the match.

Select a job posting or paste its description here...

Full report options

Generate full readiness report

Figure 4. Job discovery interface.

The job discovery screen is a core final feature because it allows users to search for jobs inside the application and select a posting for direct resume-job fitting. This reduces the need to manually copy a job description from an external site into a separate scanning tool.

9.4.5 Selected-Job Resume Comparison Preview

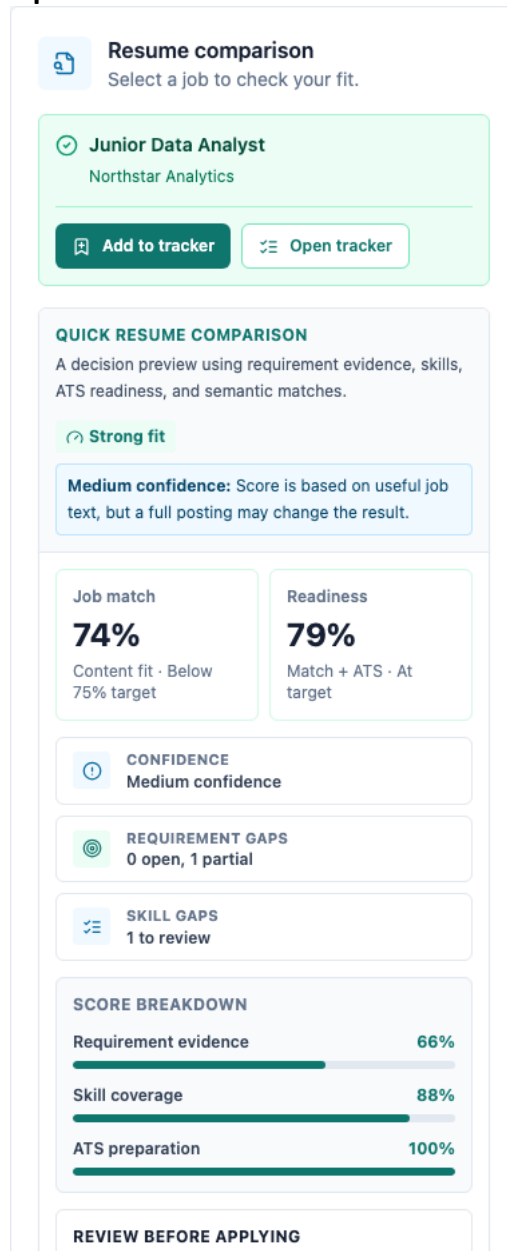


Figure 5. Selected-Job Resume Comparison Preview

9.4.6 Application Readiness Report Interface

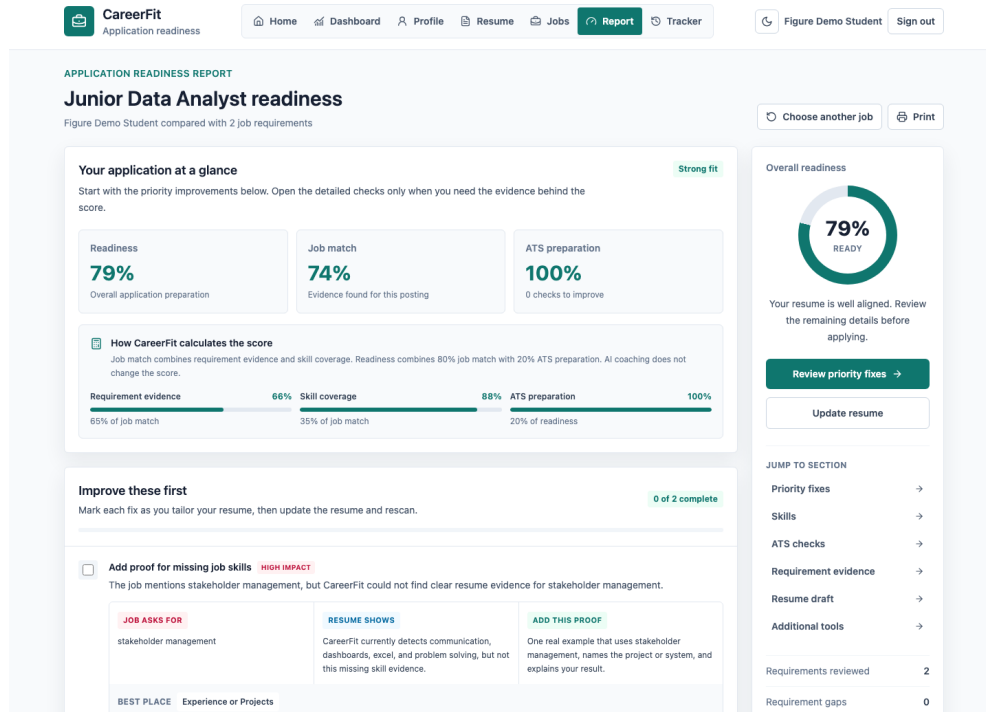


Figure 6. Application readiness report interface.

9.4.7 Application Tracker Interface

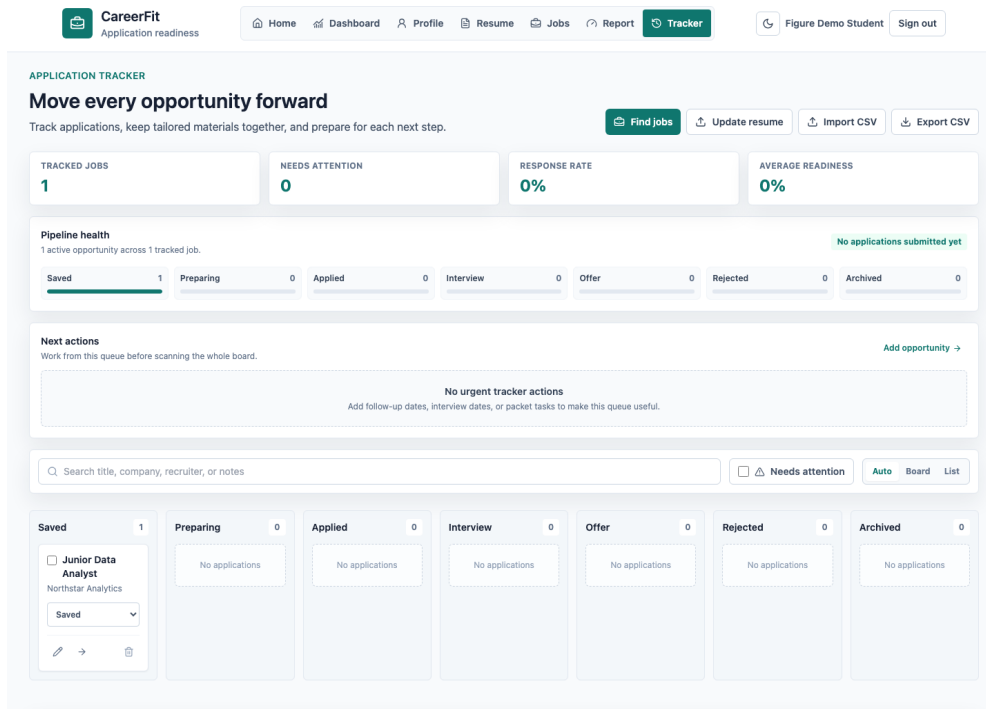


Figure 7. Application tracker interface.

These interfaces demonstrate the implemented user workflow, which guides the user from resume input to job selection, selected-job comparison, report generation, and application tracking. The final implementation includes backend-connected resume parsing, job search, matching, scoring, report generation, and tracker functionality. Optional AI coaching and embedding-based similarity remain configuration-dependent extensions.

10. User Workflow

The CareerFit workflow connects the practical application process with the proposed research matching framework.

The user first creates or updates a profile and provides a resume. Extracted resume text can be reviewed and corrected before analysis. The user then searches for a job, imports a supported posting, or provides a job description manually.

After a target job is selected, the research matching workflow begins. The job description is decomposed into individual requirements, the resume is divided into candidate evidence segments, and the CareerFit matching framework evaluates each requirement against the available resume evidence.

The user then receives an explainable readiness report showing:

- individual job requirements;
- the strongest supporting resume evidence where available;
- Matched, Partially Matched, Weakly Supported, or Missing classifications;
- job-match and ATS-preparation information; and
- recommendations linked to identified evidence gaps.

The user may then revise the resume, repeat the analysis, and save the selected position to the application tracker.

This workflow allows the implemented matching framework to be demonstrated within a realistic application-preparation process while keeping the research matching logic conceptually separate from job discovery and tracking functionality.

CareerFit User Workflow

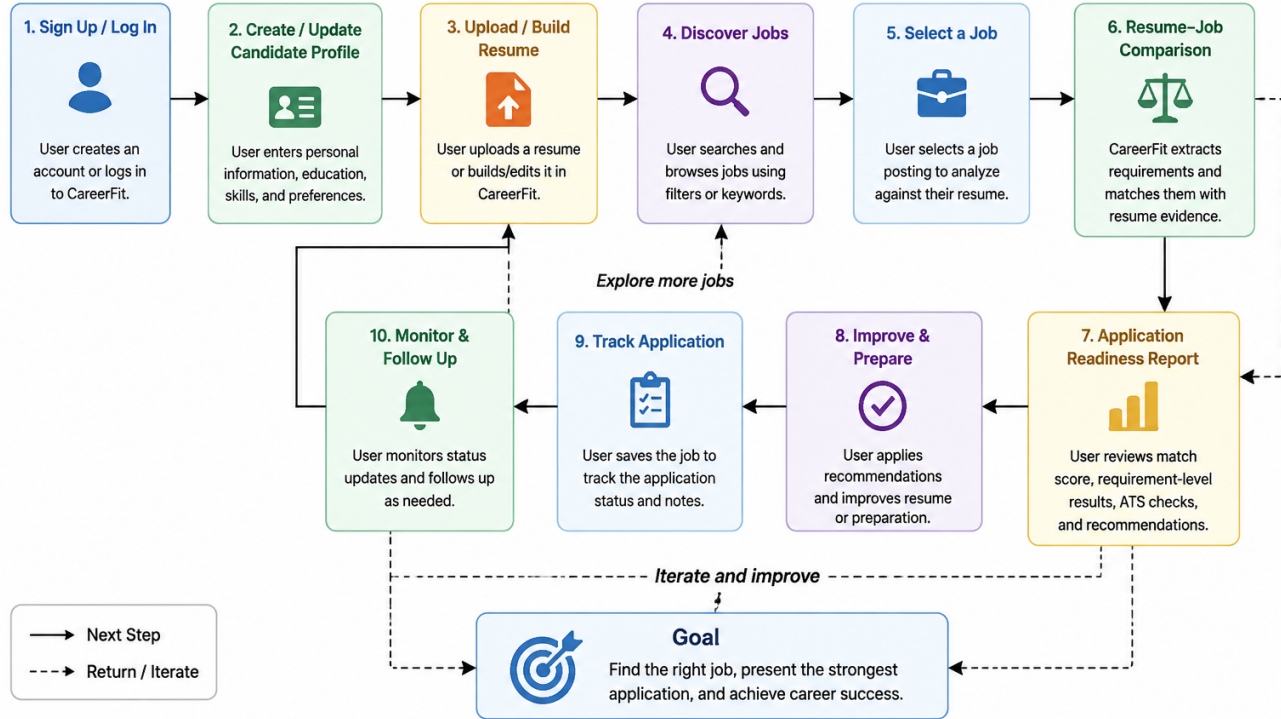


Figure 8. CareerFit user workflow. After resume input and job selection, the proposed matching framework performs requirement-level evidence analysis before the user reviews the explainable readiness report and continues to application tracking.

The main workflow value is that the user can search for jobs and directly fit the resume to a selected posting without leaving the application. The system also supports recovery paths such as manual resume text entry, pasted job descriptions, and sample jobs if external provider data is unavailable.

10.1 Alternative and Recovery Flows

Situation	Proposed Recovery
Resume extraction is incomplete	User can edit the extracted text or paste the resume manually.
Unsupported file is uploaded	System shows a clear error and allows replacement.
Job provider is unavailable	User can use another provider, import a URL, or paste the job description.
Job description is incomplete	System warns the user and suggests pasting the full posting.
User selects a job without a resume	System prompts the user to upload or paste a resume first.
AI coaching is unavailable	Deterministic matching and reporting still work.

11. Detailed System Description

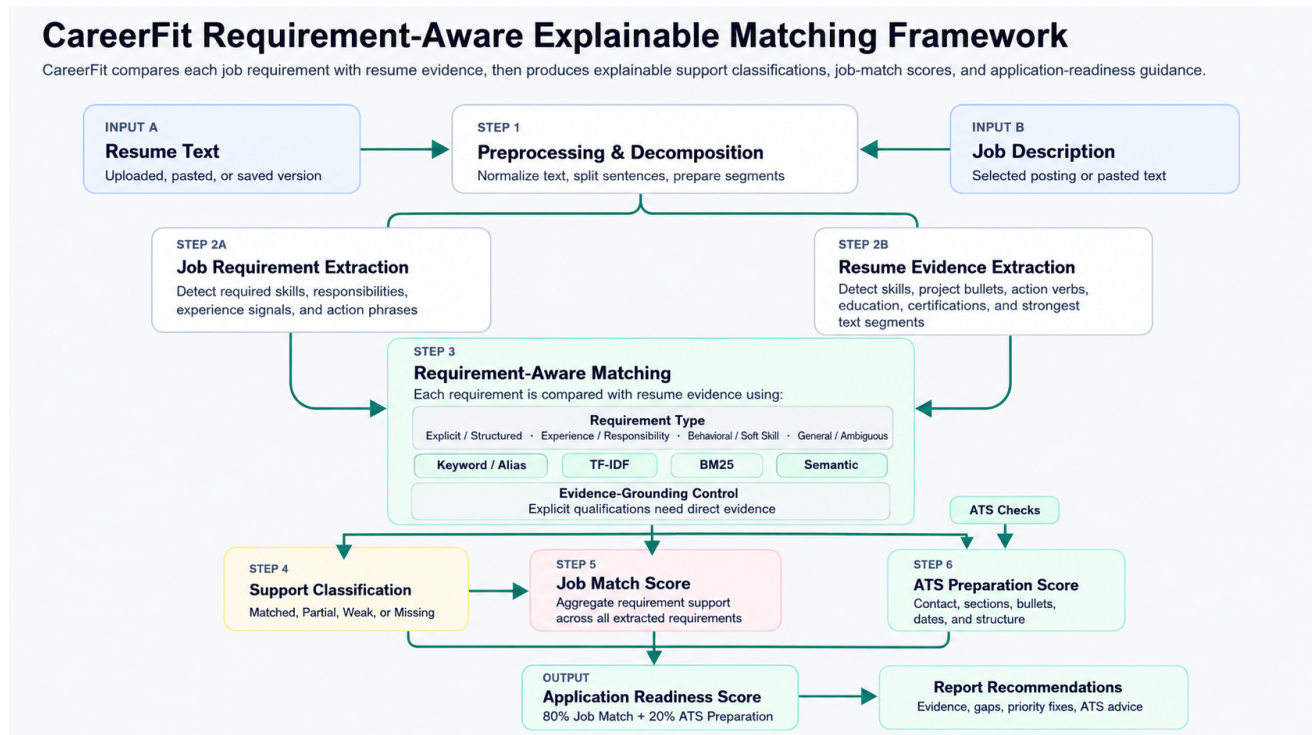


Figure 9. CareerFit Requirement-Aware Explainable Matching Framework. The framework decomposes the job description into individual requirements, identifies requirement type, evaluates candidate resume evidence using complementary lexical and semantic signals, applies requirement-dependent weighting and evidence-grounding rules, and produces explainable requirement-level decisions before calculating job-match and application-readiness scores.

The central research component of CareerFit is the CareerFit Requirement-Aware Explainable Matching Framework. The framework is designed to determine whether individual requirements in a job description are supported by specific evidence contained in a candidate's resume.

The framework differs from a whole-document similarity approach. Instead of representing the complete job description and complete resume as two documents and producing only one similarity value, CareerFit decomposes the matching problem into a collection of smaller requirement-evidence decisions.

For each extracted job requirement, the framework searches the resume for candidate evidence and evaluates the relationship using multiple complementary signals. These signals include direct keyword and skill-alias evidence, TF-IDF similarity, BM25-style lexical relevance, and semantic similarity.

The individual algorithms are not presented as original contributions. The proposed contribution is the way these signals are organized within a requirement-aware and evidence-grounded decision process.

The framework follows the general transformation:

Job Description → Individual Requirements → Requirement Type → Resume Evidence Segments → Matching Signals → Requirement Support Score → Support Classification → Explanation

The purpose of this design is to investigate whether different types of job requirements should be evaluated differently and whether requiring explicit supporting evidence can provide a more transparent resume-job assessment.

11.1 Requirement-Level Unit of Analysis

CareerFit performs matching at the level of **individual job requirements and resume evidence**, rather than relying only on a single similarity score between the complete resume and the complete job description.

The job description is first decomposed into individual requirements. These requirements may refer to technical skills, tools, responsibilities, experience, education, certifications, or behavioral competencies.

The resume is then divided into meaningful evidence segments, such as:

- experience bullet points;
- project descriptions;
- listed technical skills;
- education entries;
- certification entries; and
- other statements describing relevant achievements or responsibilities.

For each job requirement, CareerFit searches these resume segments and identifies the strongest available supporting evidence.

The matching process can therefore be represented conceptually as:

Job Requirement → Candidate Resume Evidence → Matching Evaluation → Support Decision

For example:

Job requirement:

“Experience developing REST APIs.”

Possible resume evidence:

“Developed RESTful backend services using Django REST Framework.”

CareerFit evaluates the relationship between these two pieces of information and determines how strongly the resume supports the requirement.

A second example illustrates why requirement-level analysis is useful:

Job requirement:

“Ability to collaborate with cross-functional teams.”

Possible resume evidence:

“Worked with product, support, and development teams to deliver customer-facing features.”

Although the wording is different, the resume statement may still provide meaningful evidence for the requirement. A method based only on exact keyword overlap may fail to recognize this relationship, while lexical-semantic matching may identify it.

The important distinction is that CareerFit does not assume that a high overall similarity between a resume and a job description means that every individual requirement is satisfied.

For example, a resume may be highly aligned with a software engineering position overall but contain no evidence of a required AWS certification. A whole-document similarity score could hide this gap. Requirement-level analysis allows CareerFit to identify the certification separately and classify it as unsupported.

For every requirement, the framework therefore produces three main outputs:

Output	Description
Supporting Evidence	The resume segment that most strongly supports the requirement, where evidence is available.
Support Score	A numerical assessment calculated from the relevant matching signals.
Support Classification	The final interpretation: Matched, Partially Matched, Weakly Supported, or Missing.

This requirement-level structure forms the foundation of the proposed CareerFit framework. It allows the system to explain not only **how well** a resume fits a job, but also **which specific requirements are supported, what evidence supports them, and where important gaps remain**.

11.2 Requirement and Resume-Evidence Decomposition

The first stage of the framework transforms unstructured job and resume text into smaller analysis units.

The job description is divided into candidate requirements using sentence boundaries, bullet points, headings, punctuation, and requirement-related patterns. Requirements may describe technical skills, software tools, qualifications, responsibilities, experience expectations, certifications, education, or behavioral capabilities.

The resume is independently divided into evidence segments. Resume headings such as **Experience**, **Projects**, **Skills**, **Education**, and **Certifications** provide contextual information that helps preserve the meaning of each segment.

For example, consider the following job requirement:

“Experience developing REST APIs.”

Possible resume evidence may include:

“Developed RESTful backend services using Django REST Framework.”

or:

“Built backend endpoints and integrated third-party services for a web application.”

The first example contains strong direct lexical evidence. The second contains less direct vocabulary overlap but may still represent relevant experience.

CareerFit therefore does not assume that one similarity technique is appropriate for every requirement-evidence relationship.

11.3 Requirement-Aware Matching

A central design decision in the proposed framework is that job requirements are not treated as homogeneous.

Different types of requirements provide different forms of evidence. CareerFit therefore assigns each extracted requirement to a broad requirement category before calculating its final support score.

Requirement Category	Example	Most Important Evidence Characteristics
Explicit / Structured	“Python”, “AWS certification”, “Bachelor's degree in Computer Science”	Exact terminology, recognized aliases, structured resume sections
Experience / Responsibility	“Develop REST APIs”, “manage software projects”, “design scalable services”	Lexical similarity, related terminology, semantic evidence, experience statements
Behavioral / Soft Skill	“Cross-functional collaboration”, “strong communication skills”	Contextual and semantic evidence, descriptions of actions and collaboration
General / Ambiguous	Requirements that cannot be confidently assigned to another category	Balanced use of available matching signals

Requirement typing allows the framework to vary the importance of different matching signals.

For an explicit technical requirement such as **“Python”**, exact terminology or a recognized alias provides particularly strong evidence. Semantic similarity should not override the absence of the required skill merely because a resume discusses generally related software-development concepts.

For a responsibility such as **“develop REST APIs,”** lexical similarity remains useful, but related wording such as **“implemented backend endpoints”** may also represent meaningful evidence. Semantic information therefore has a larger role.

For a behavioral requirement such as **“collaborate with cross-functional teams,”** exact keyword matching is less reliable because resumes often describe the behavior rather than repeat the

employer's terminology. Evidence such as **“worked with product, support, and development teams”** should therefore receive greater semantic consideration.

The requirement category is consequently used to determine how the individual matching signals are combined.

This creates the following decision process:

Requirement Type → Matching-Signal Weights → Requirement-Evidence Score

In other words, the type of job requirement determines the relative importance assigned to the available matching signals before the final requirement-evidence score is calculated.

The hypothesis investigated by CareerFit is that this **requirement-aware weighting strategy** can provide more appropriate requirement-evidence decisions than applying the same global similarity rule to every type of job requirement.

The exact scoring model and signal weights are defined in the following subsections.

11.4 Text Preprocessing and Matching Signals

Before matching is performed, both job requirements and resume evidence segments are normalized so that the different matching methods operate on comparable text.

The proposed preprocessing pipeline includes:

- conversion to lowercase;
- removal of unnecessary URLs and source metadata;
- tokenization;
- punctuation normalization;
- removal of common stop words where appropriate;
- normalization of common plural forms;
- normalization of known technical-skill aliases; and
- preservation of meaningful resume-section context such as Experience, Projects, Skills, Education, and Certifications.

Full stemming or lemmatization is not required for the initial CareerFit implementation because excessive word reduction may reduce the interpretability of technical terminology. Instead, the framework uses controlled normalization and explicit aliases for common skills and technologies.

For each job requirement and candidate resume-evidence segment, CareerFit calculates four normalized matching signals between 0 and 1.

Signal	Description
Keyword/Alias Signal (K)	Measures direct keyword overlap and recognized skill aliases between the requirement and resume evidence.
TF-IDF Signal (T)	Measures weighted lexical similarity using TF-IDF cosine similarity.

BM25 Signal (B)	Measures lexical relevance using BM25-style ranking between the requirement and candidate evidence segments.
Semantic Signal (S)	Measures related meaning when the requirement and resume evidence use different wording.

Each signal captures a different form of evidence.

Keyword and alias matching is particularly useful for explicit skills such as Python, React, AWS, or PostgreSQL. TF-IDF and BM25 provide broader lexical evidence by considering important shared vocabulary. TF-IDF and BM25 are established information-retrieval techniques for weighting and ranking textual evidence based on term importance and relevance [1], [2], [21]–[23]. Semantic matching can also be supported by distributed word representations or sentence-level embedding models such as Sentence-BERT [4], [5].

The proposed framework does not assume that these four signals should contribute equally to every type of requirement. Their relative importance is determined by the requirement-aware weighting mechanism described in the next section.

11.5 Requirement-Aware Scoring Model

The proposed CareerFit scoring model combines the four matching signals according to the type of job requirement being evaluated.

The main hypothesis is that a single fixed combination of matching signals is not equally appropriate for every requirement. Explicit technical requirements should rely more strongly on direct terminology and recognized aliases, whereas responsibilities and behavioral requirements may require stronger semantic interpretation.

For each requirement-evidence pair, CareerFit calculates:

$$\text{Requirement-Evidence Score} = wK(K) + wT(T) + wB(B) + wS(S)$$

where:

- **K** = keyword and alias score;
- **T** = TF-IDF similarity score;
- **B** = normalized BM25 relevance score;
- **S** = semantic similarity score; and
- **wK, wT, wB, and wS** are requirement-dependent weights.

All four matching signals are normalized to the range **0 to 1**, and the resulting score is converted to a percentage between **0 and 100**.

11.5.1 Requirement-Dependent Signal Weights

CareerFit uses different initial weights according to the requirement category.

Requirement Type	Keyword / Alias	TF-IDF	BM25	Semantic	Rationale
Explicit / Structured	45%	20%	20%	15%	Direct terminology and recognized aliases provide the strongest evidence for explicit skills, education, and certifications.
Experience / Responsibility	15%	25%	25%	35%	Responsibilities may be expressed using related but non-identical wording, so semantic evidence receives greater importance.
Behavioral / Soft Skill	10%	20%	20%	50%	Behavioral evidence is often described through actions rather than exact terminology.
General / Ambiguous	25%	25%	25%	25%	No signal is assumed to be dominant when the requirement type cannot be confidently identified.

These weights are explicit design parameters rather than learned parameters. They were defined before the reported controlled evaluation and were not adjusted to fit the final reported test outcomes. Because the current evaluation is small, a broader sensitivity analysis examining reasonable variations in these weights remains future work.

11.5.2 Evidence-Grounding Constraint

CareerFit introduces an additional constraint for explicit or structured requirements.

A high semantic similarity score alone should not be sufficient to claim that an explicit skill, degree, or certification is present.

For example, if a job explicitly requires:

“AWS Certified Solutions Architect”

a resume should not be classified as having matched that requirement merely because it contains generally related cloud-computing experience.

Therefore, for explicit or structured requirements, the absence of direct terminology, a recognized alias, or appropriate structured evidence prevents the requirement from receiving the highest support classification.

This rule is intended to reduce semantic false positives while preserving the ability of semantic matching to assist with less structured responsibilities and behavioral requirements.

11.5.3 Selection of Supporting Resume Evidence

Each job requirement is compared against all relevant resume evidence segments.

CareerFit calculates a requirement-evidence score for each candidate segment and selects the segment with the strongest valid score as the primary supporting evidence.

The process is therefore:

Job Requirement → Candidate Resume Segments → Score Each Segment → Select Strongest Evidence → Assign Support Classification

The selected evidence is stored with the result so that the user can inspect the resume statement responsible for the decision.

If no candidate evidence reaches the minimum support threshold, the requirement is classified as **Missing**, and no unsupported evidence is presented as justification.

11.5.4 Requirement Support Classification

After the strongest evidence has been selected, the requirement-evidence score is converted into an interpretable support category.

Score	Classification	Interpretation
70–100	Matched	The resume provides strong and relevant evidence for the requirement.
40–69	Partially Matched	Relevant evidence exists, but the requirement is not fully demonstrated.
20–39	Weakly Supported	Some related evidence exists, but it is indirect or insufficient.
0–19	Missing	No meaningful supporting evidence was identified.

These categories are intended to make the numerical score understandable rather than presenting the score as a prediction of hiring success.

The threshold values are explicit design parameters of the framework and were kept fixed during the reported controlled comparison. Their suitability remains a limitation of the current study, and a broader threshold-sensitivity analysis is identified as future work.

11.5.5 Job-Level Match Score

The final job-match score is calculated only after the individual requirements have been evaluated.

Required qualifications and preferred qualifications may have different importance. CareerFit therefore assigns an importance factor to each requirement:

- **Required / mandatory requirement:** weight = 1.0
- **Preferred / desirable requirement:** weight = 0.6
- **Unspecified requirement:** weight = 1.0

The job-match score is calculated as the weighted average of the individual requirement support scores.

This structure prevents a high whole-document similarity from hiding important missing requirements.

For example, a candidate may strongly match eight requirements but completely lack one mandatory certification. CareerFit retains the missing certification as an explicit requirement-level result rather than allowing it to disappear inside a single similarity calculation.

11.5.6 Separation of Match Score and Application-Readiness Score

CareerFit distinguishes between **job-content alignment** and **resume preparation quality**.

The **Job Match Score** measures how strongly the resume provides evidence for the requirements of the selected position.

The **ATS Preparation Score** evaluates separate resume-preparation characteristics such as structural completeness and common ATS-oriented checks.

These scores should first be displayed independently because they represent different concepts.

For the overall application-readiness indicator, CareerFit uses:

Application Readiness Score = 80% Job Match Score + 20% ATS Preparation Score

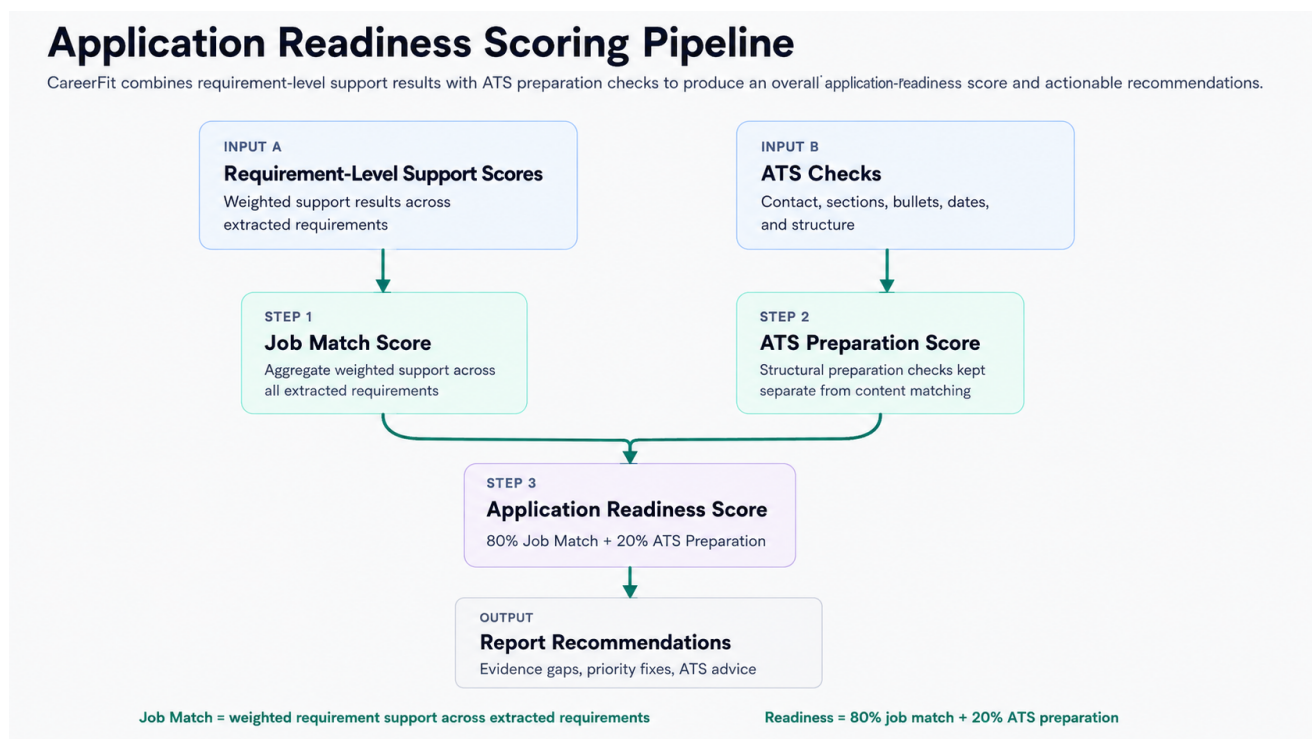


Figure 10. Application Readiness Scoring Pipeline. CareerFit calculates the Job Match Score by aggregating weighted requirement-level support scores and calculates the ATS Preparation Score independently from structural resume checks. The two scores are then combined using 80% Job Match and 20% ATS Preparation to produce the overall Application Readiness Score and related recommendations.

This weighting intentionally gives greater importance to actual evidence of job requirements than to formatting or preparation checks.

The application-readiness score is not interpreted as the probability of receiving an interview, an offer, or any other hiring outcome.

11.6 Semantic Evidence and False-Positive Control

Semantic matching is used in CareerFit to identify relevant resume evidence when the wording of a job requirement differs from the wording used in the resume.

For example, the requirement:

“Collaborate with cross-functional teams”

may be supported by the resume statement:

“Worked with product, development, and support teams to deliver new features.”

A purely lexical method may assign a low similarity score because the two statements share limited vocabulary. Semantic evidence can help recognize that the underlying experience is related.

However, semantic similarity is not treated as sufficient evidence for every type of requirement.

For explicit requirements such as technical skills, degrees, licenses, or certifications, CareerFit applies the evidence-grounding constraint defined in Section 11.5.2. A high semantic similarity score cannot by itself establish that the candidate possesses an explicitly required qualification.

For example:

Job requirement: “AWS Certified Solutions Architect”

Resume evidence: “Developed cloud-based distributed applications”

Although these statements are semantically related to cloud computing, the resume statement does not demonstrate possession of the required certification. CareerFit should therefore avoid classifying the requirement as fully matched without explicit certification evidence.

Semantic matching consequently has two roles within the proposed framework:

- increasing recall for responsibilities and behavioral requirements that may be expressed through paraphrased language; and
- complementing, rather than replacing, direct evidence for explicit and structured requirements.

The implemented semantic component uses controlled semantic concept mappings and can optionally be extended with embedding-based similarity. The contribution of the overall CareerFit framework does not depend on a particular embedding model. Instead, semantic similarity is treated as one evidence signal whose influence depends on the type of requirement being evaluated.

This design allows semantic matching to improve recognition of paraphrased evidence while reducing the risk that general semantic similarity produces unsupported strong matches.

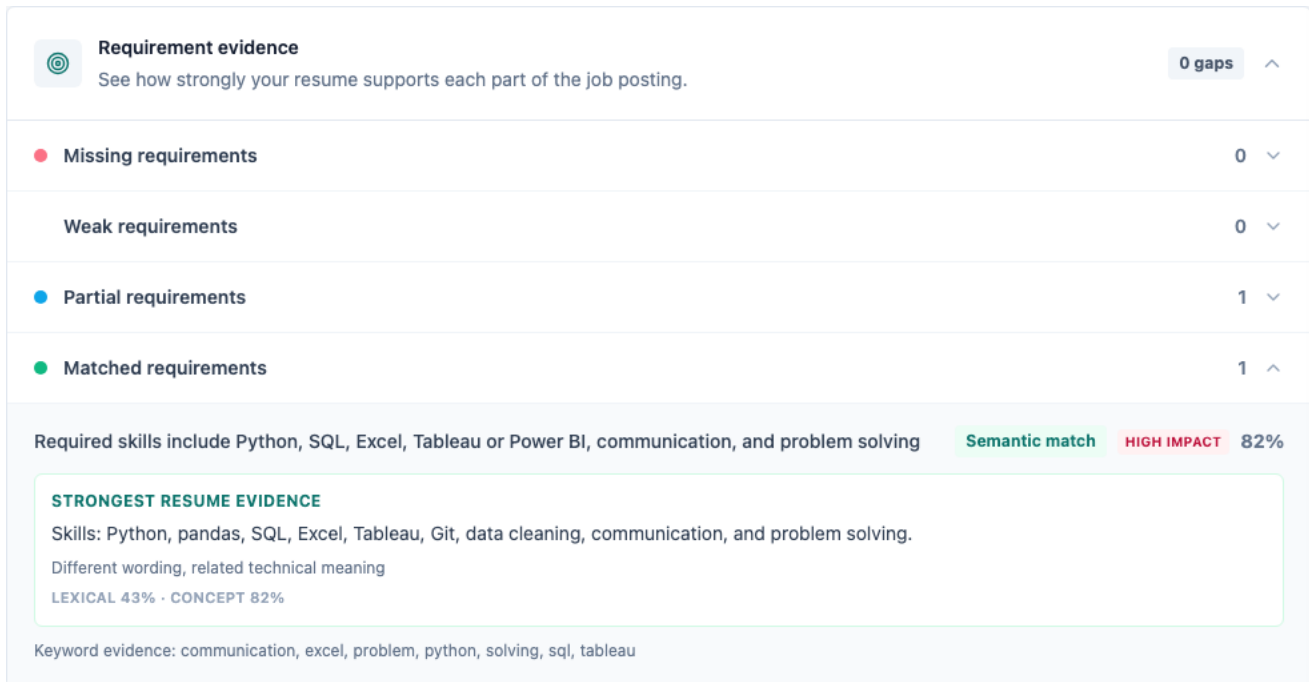


Figure 11. Example of requirement-level evidence grounding in CareerFit. The system links an extracted job requirement to the strongest supporting resume segment and displays the evidence used to justify the resulting support classification.

Figure 11 illustrates how the proposed framework exposes the evidence behind a requirement-level decision. Rather than presenting only a similarity score, CareerFit identifies the resume segment selected as supporting evidence and makes this evidence visible to the user.

11.7 Explainable Recommendations

CareerFit recommendations are derived from the results of the requirement-level analysis rather than generated independently from the matching process.

The purpose of the recommendation component is to translate identified evidence gaps into practical actions for the user.

Recommendations may be generated when:

- a required skill is missing;
- a job requirement is classified as Missing or Weakly Supported;
- a requirement is only Partially Matched and stronger resume evidence may be needed;
- relevant experience exists but is not clearly expressed in the resume; or
- an ATS-oriented preparation check identifies a structural issue.

For example, if a requirement is classified as **Partially Matched** because relevant experience is present but the evidence is indirect, CareerFit may recommend rewriting the relevant resume bullet to make the connection to the job requirement clearer.

If no supporting evidence is found for a required technical skill, the system should not recommend falsely adding that skill. Instead, it should identify the requirement as a genuine gap.

This distinction is important because CareerFit is intended to support accurate resume improvement rather than encourage users to claim experience they do not possess.

The core recommendation process is deterministic and directly connected to the requirement classifications and ATS checks. Optional AI-assisted coaching may be added as a secondary feature, but it remains separate from the deterministic matching and scoring framework.

This preserves the explainability of the research component because every core recommendation can be traced back to a specific requirement, evidence gap, or preparation issue.

11.8 CareerFit Matching Algorithm

The complete CareerFit matching procedure combines the previous components into a single requirement-level decision process.

The algorithm receives two primary inputs:

- the normalized resume text; and
- the selected job description.

It produces:

- an evidence-backed result for each job requirement;
- a requirement support score;
- a support classification;
- a job-match score;
- an ATS preparation score; and
- an overall application-readiness score.

The implemented procedure is summarized in Algorithm 1.

Algorithm 1 — CareerFit Requirement-Aware Explainable Matching

Input: Resume (CV), Job Description (JD)

Output: Requirement-level evidence mappings, support classifications, Job Match Score, and Application Readiness Score

Step 1-Preprocess the inputs

Normalize the resume and job-description text while preserving meaningful technical terminology and resume-section information.

Step 2-Extract job requirements

Decompose the job description into individual requirements, including technical skills, qualifications, responsibilities, experience expectations, certifications, and behavioral requirements.

Step 3-Extract resume evidence

Divide the resume into meaningful evidence segments such as experience bullets, project descriptions, skill entries, education information, and certifications.

Step 4-Determine the requirement type

For each extracted job requirement, assign one of the following categories:

- Explicit / Structured;
- Experience / Responsibility;
- Behavioral / Soft Skill; or
- General / Ambiguous.

Step 5-Generate candidate evidence

For each requirement, identify the resume segments that may provide relevant supporting evidence.

Step 6-Calculate matching signals

For every candidate requirement-evidence relationship, calculate:

- Keyword/Alias score;
- TF-IDF similarity;
- BM25 relevance; and
- Semantic similarity.

Each signal is normalized to a common scale.

Step 7-Apply requirement-aware weighting

Select the matching-signal weights associated with the requirement type and calculate the combined requirement-evidence score.

For example, explicit technical requirements place greater emphasis on direct keyword and alias evidence, while behavioral requirements place greater emphasis on semantic evidence.

Step 8-Apply evidence-grounding rules

For explicit qualifications such as required technologies, degrees, certifications, or licenses, prevent semantic similarity alone from producing a strong match when direct evidence is absent.

Step 9-Select the strongest evidence

Compare the scores of the candidate resume segments and select the strongest valid supporting segment for the requirement.

Step 10-Assign the support classification

Convert the strongest valid requirement-evidence score into one of four categories:

- **Matched:** 70–100
- **Partially Matched:** 40–69
- **Weakly Supported:** 20–39
- **Missing:** 0–19

Store both the classification and the supporting evidence.

Step 11-Repeat for all requirements

Repeat Steps 4–10 independently for every extracted job requirement.

Step 12-Calculate the Job Match Score

Aggregate the individual requirement scores using their importance weights. Required requirements receive greater importance than preferred requirements.

Step 13-Calculate ATS Preparation

Evaluate ATS-oriented preparation checks separately from the job-content matching process.

Step 14-Calculate Application Readiness

Combine the two dimensions using:

$$\text{Application Readiness Score} = 0.80 \times \text{Job Match Score} + 0.20 \times \text{ATS Preparation Score}$$

Step 15-Generate explanations and recommendations

Present:

- each extracted job requirement;
- its support classification;
- the strongest supporting resume evidence;
- missing or weak areas;
- ATS preparation issues; and
- recommendations linked to the identified gaps.

11.8.1 Algorithm Design Rationale

The CareerFit algorithm contains three design decisions that distinguish it from applying a single similarity method to the full resume and job description.

First, matching is performed at the **requirement level**. This prevents strong similarity in one area of a resume from hiding a missing requirement elsewhere.

Second, the algorithm is **requirement-aware**. The same combination of matching signals is not applied uniformly to technical skills, experience statements, certifications, and behavioral requirements.

Third, every positive matching decision is designed to be **evidence-grounded**. The framework attempts to identify the resume statement responsible for the decision so that the user can inspect and understand the result.

These three properties form the methodological core of CareerFit and provide the basis for the controlled evaluation described in Section 12.

12. Research Validation and Experimental Methodology

CareerFit Research Evaluation Pipeline

CareerFit evaluates the same labelled requirement-evidence cases across baseline methods and the full framework using classification and evidence-retrieval measures.

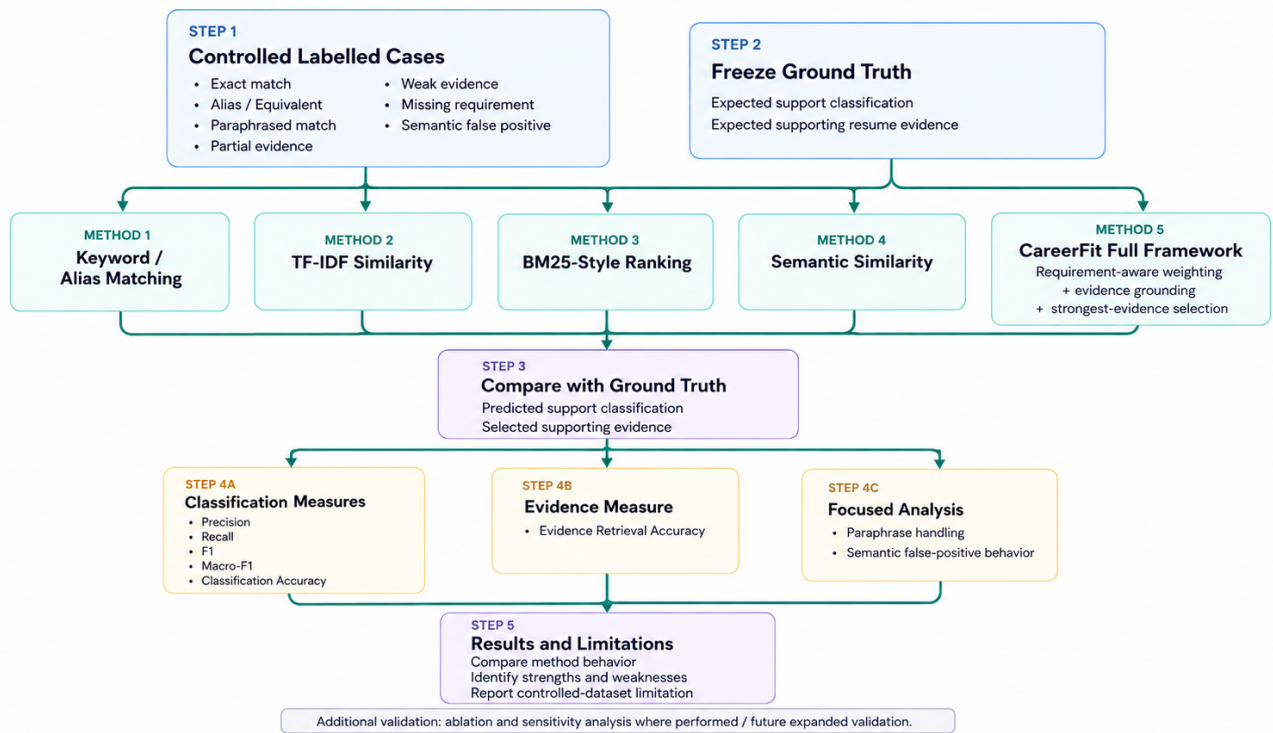


Figure 12. CareerFit Research Evaluation Pipeline. The same manually labelled requirement-evidence cases are evaluated using keyword/alias matching, TF-IDF similarity, BM25-style lexical ranking, semantic similarity, and the complete CareerFit Requirement-Aware Explainable Matching Framework. Predicted support classifications and selected resume evidence are compared with frozen ground truth using classification and evidence-retrieval measures, with additional analysis of paraphrased evidence and semantic false-positive cases.

Figure 12 presents the broader evaluation design defined for CareerFit. The completed quantitative results reported in Section 13 cover the experiments for which outputs were recorded: the controlled skill-matching evaluation and the five-case paraphrase-ranking comparison of keyword overlap, TF-IDF, and the completed CareerFit framework. BM25, semantic-only comparison, expanded classification evaluation, sensitivity analysis, and ablation remain part of broader future validation.

The purpose of the CareerFit evaluation is to examine the matching behavior and explainability of the implemented requirement-aware framework under controlled conditions. The broader evaluation

methodology defines comparisons with keyword/alias matching, TF-IDF, BM25-style ranking, and semantic similarity, while the completed quantitative results reported in Section 13 are limited to the experiments that were executed and recorded.

Software correctness and research validity are evaluated separately. Functional tests verify that the web application operates correctly, while the research evaluation examines the behavior and effectiveness of the matching framework itself.

The research validation is designed around manually labelled requirement-evidence cases so that each matching method can be evaluated against the same expected result.

The evaluation addresses three questions derived from the research questions defined in Section 2.2:

EQ1: How accurately do the individual baseline methods and the proposed CareerFit framework classify job requirements as Matched, Partially Matched, Weakly Supported, or Missing?

EQ2: How effectively does each method identify the correct supporting resume evidence when job requirements and resume statements use direct versus paraphrased wording?

EQ3: Does requirement-aware weighting and evidence grounding improve the balance between correct matching and explainability compared with applying the individual matching techniques independently?

12.1 Evaluation Dataset and Ground-Truth Construction

The CareerFit evaluation methodology is based on controlled resume-job requirement cases created specifically for requirement-level analysis. The completed experiments reported in Section 13 use a 10-case controlled skill-matching set and five controlled paraphrase-ranking cases, while the broader evaluation design defines additional case categories for expanded future validation.

Each evaluation case contains:

- one job requirement;
- a set of candidate resume evidence segments;
- the manually identified strongest supporting evidence, where such evidence exists;
- the expected support classification; and
- the requirement category used by the proposed framework.

The expected labels are defined manually before the matching methods are evaluated. This manually defined reference is treated as the **ground truth** for the controlled experiment.

The evaluation cases should represent different forms of resume-job relationships rather than only exact keyword matches.

The broader evaluation design includes the following case categories:

Case Category	Purpose
Exact Match	Tests cases where the resume and requirement use the same or directly normalized terminology.

Alias / Equivalent Term	Tests recognized technical equivalents such as React.js / React or RESTful / REST API.
Paraphrased Match	Tests cases where the same experience is expressed using different vocabulary.
Partial Evidence	Tests cases where the resume supports only part of the requirement.
Weak Evidence	Tests related but indirect evidence that should not be considered a strong match.
Missing Requirement	Tests requirements for which the resume provides no relevant evidence.
Semantic False-Positive Case	Tests semantically related text that should not be accepted as proof of an explicit qualification.

For example, the following case evaluates paraphrased behavioral evidence:

Job requirement:

“Collaborate effectively with cross-functional teams.”

Expected resume evidence:

“Worked with product, development, and support teams to deliver customer-facing features.”

The correct result should identify the resume statement as relevant evidence even though there is limited exact vocabulary overlap.

A contrasting false-positive case is:

Job requirement:

“AWS Certified Solutions Architect.”

Resume statement:

“Developed distributed cloud-based applications.”

The statements are related to the same technical domain, but the resume statement does not demonstrate possession of the certification. The expected classification should therefore not be Matched.

Including both positive and negative cases is important because evaluation should measure not only whether CareerFit finds additional semantic relationships, but also whether it avoids unsupported matches.

12.2 Baseline Methods

The broader comparative methodology defines four individual matching baselines that can be evaluated under the same controlled conditions as the complete CareerFit framework.

The baselines are:

Baseline 1 — Keyword and Alias Matching

This baseline uses exact terminology, normalized tokens, and known technical aliases. It provides a highly interpretable reference method and is expected to perform strongly on explicit technical requirements.

Baseline 2 — TF-IDF Cosine Similarity

This baseline calculates weighted lexical similarity between each requirement and resume evidence segment. It represents a traditional information-retrieval approach that considers term importance but remains dependent on shared vocabulary.

Baseline 3 — BM25-Style Lexical Ranking

This baseline ranks candidate resume evidence using BM25-style lexical relevance. It provides a second information-retrieval baseline for evaluating whether ranked lexical matching improves evidence retrieval.

Baseline 4 — Semantic Similarity

This baseline uses semantic similarity independently of the proposed requirement-aware weighting mechanism. Its purpose is to measure the benefit and risk of relying primarily on meaning-based similarity.

Proposed Method — CareerFit Requirement-Aware Explainable Matching

In the completed five-case paraphrase-ranking experiment reported in Section 13.2, quantitative results were recorded for keyword overlap, TF-IDF cosine similarity, and the completed CareerFit framework. BM25 and semantic-only matching remain defined baselines for an expanded comparison, but numerical results for those methods are not reported in the current evaluation.

The complete CareerFit framework combines the available evidence signals according to requirement type, applies evidence-grounding constraints, selects the strongest resume evidence, and produces an interpretable support classification.

Whenever methods are directly compared, they receive the same requirements, candidate evidence segments, and manually defined expected outputs so that differences in results are attributable to the matching approach rather than different input data.

12.3 Evaluation Metrics

The broader CareerFit evaluation methodology defines measures at two levels: requirement classification and evidence retrieval. The completed experiments reported in Section 13 use the metrics applicable to the recorded test sets, while additional classification measures remain available for expanded validation.

12.3.1 Requirement Classification Metrics

For requirement-level classification, the predicted category is compared with the manually defined expected category.

The following measures are defined for requirement-level classification evaluation:

- **Precision:** measures how often predicted matches are correct.
- **Recall:** measures how many expected matches are successfully identified.
- **F1-score:** combines precision and recall into a single measure.
- **Classification Accuracy:** measures the proportion of requirements assigned to the correct support category.
- **Macro F1-score:** gives equal importance to the Matched, Partially Matched, Weakly Supported, and Missing classes.

Precision, recall, F1-score, and related classification measures are widely used for evaluating classification systems and provide complementary views of predictive performance [24], [25].

Macro F1 is useful because the evaluation should not appear successful simply because one support category occurs more frequently than the others.

The completed five-case paraphrase experiment is an evidence-ranking comparison rather than a complete four-class classification experiment; therefore, classification accuracy and Macro F1 are not reported as final numerical results for that experiment.

12.3.2 Evidence Retrieval Metrics

Correct classification alone is not sufficient for an explainable matching system. CareerFit should also identify the correct resume evidence behind its decision.

Evidence retrieval is therefore evaluated by checking whether the resume segment selected by the method matches the manually identified supporting evidence.

The primary evidence measure is:

Evidence Retrieval Accuracy = Number of requirements for which the correct supporting segment is retrieved / Number of requirements with expected supporting evidence

This measure directly evaluates the explainability objective of the proposed framework.

12.3.3 Paraphrase-Specific Evaluation

A paraphrase-specific analysis was performed because one of the main weaknesses of lexical matching is its dependence on shared vocabulary.

The completed experiment used five controlled cases in which the job requirement and correct resume evidence expressed related meaning using different wording. Keyword overlap, TF-IDF cosine similarity, and the completed CareerFit framework were evaluated on the same five cases.

The recorded results are reported in Section 13.2. Keyword overlap and TF-IDF each retrieved the expected evidence in 2 out of 5 cases, while CareerFit retrieved the expected evidence in 5 out of 5 cases. BM25 and semantic-only matching remain part of the broader comparative methodology but are not assigned numerical results in the current report.

12.4 Experimental Procedure

The completed controlled evaluation followed a fixed procedure so that the compared methods operated on the same manually defined cases.

First, the expected supporting evidence was defined before the matching methods were run. For the controlled skill-matching evaluation, 10 manually defined cases were used to verify the supporting extraction and matching rules.

For the paraphrase-ranking comparison, five controlled cases were prepared. Each case contained a job requirement, the expected relevant resume evidence, and an unrelated alternative segment.

Keyword overlap, TF-IDF cosine similarity, and the completed CareerFit Requirement-Aware Explainable Matching Framework were then run on the same five paraphrase cases. Each method was evaluated according to whether it ranked the manually expected resume evidence above the unrelated alternative.

The recorded outcomes were then compared with the frozen expected evidence. The resulting ranking accuracy is reported in Section 13.2.

The broader methodology also defines BM25, semantic-only matching, requirement-classification metrics, additional evidence-retrieval evaluation, sensitivity analysis, and ablation experiments. These components provide a structure for expanded validation but are not assigned unreported numerical results in the current study.

12.5 Ablation Analysis

Ablation analysis is defined as an additional validation step for investigating which components contribute to the behavior of the CareerFit framework. A complete numerical ablation study was not part of the reported controlled results and therefore remains an extension of the current evaluation.

Future ablation experiments should evaluate the following variants:

- Full CareerFit framework
- CareerFit without semantic similarity
- CareerFit without requirement-aware weighting
- CareerFit without keyword/alias normalization
- CareerFit without the evidence-grounding constraint

The purpose of this future ablation analysis would be to determine whether specific design decisions contribute meaningfully to requirement classification, paraphrase recognition, evidence retrieval, or false-positive control.

For example, removing the evidence-grounding constraint may increase semantic matches while also increasing false positives for explicit certifications or qualifications. Such a result would provide evidence for the value of the grounding mechanism even if overall recall increases.

12.6 Validation Strategy and Threats to Validity

The evaluation methodology is designed to reduce several sources of bias.

First, the expected classifications and evidence mappings are defined before the final model comparison. This reduces the risk of changing the ground truth to favor the proposed method.

Second, methods that are directly compared are evaluated using the same cases. In the completed paraphrase-ranking experiment, keyword overlap, TF-IDF, and CareerFit were evaluated on the same five controlled cases.

Third, the requirement-aware weights and classification thresholds were fixed before the reported controlled evaluation rather than repeatedly adjusted against the reported test outcomes. However, the evaluation also has important limitations.

The controlled dataset is small compared with large-scale recruitment datasets. Manual labels may contain subjective judgment, particularly when distinguishing between partially matched and weakly supported evidence. The selected resumes and requirements may also not represent every occupation or writing style.

Therefore, the experimental results should be interpreted as evidence about the behavior of the framework under the controlled evaluation conditions, not as proof that CareerFit universally outperforms all resume-job matching systems.

A larger evaluation with multiple independent annotators, additional job domains, and real-world resume-job datasets would be required for stronger generalization claims.

12.7 Software Verification

Software testing is reported separately from the research evaluation because functional correctness does not by itself demonstrate the effectiveness of the proposed matching framework.

The purpose of software verification is to confirm that the CareerFit prototype performs its intended operations reliably, including resume processing, job handling, matching requests, report generation, and user-interface workflows.

The existing prototype produced the following software-verification results:

Test Area	Verification Method	Result	Purpose
Backend	Django automated test suite	88 tests passed in the reported test run	Verifies API, resume, job, account, matching, and workflow behavior
Frontend linting	npm run lint	Passed	Checks frontend code quality and consistency
Frontend unit tests	npm run test	37 tests across 9 test files passed	Verifies interface and service behavior
Frontend build	npm run build	Passed	Confirms that the application can be built successfully
End-to-end workflow	npm run test:e2e	3 tests passed	Verifies selected user workflows across frontend and backend

These results provide evidence that the implemented prototype is technically functional under the tested conditions.

However, they should not be interpreted as evidence that the CareerFit matching framework is more accurate than alternative resume-job matching methods. Algorithmic effectiveness is evaluated separately through the experimental methodology described in Sections 12.1–12.6 and the controlled framework results reported in Sections 13.1–13.3.

13. Evaluation Results and Discussion

Following implementation of the CareerFit Requirement-Aware Explainable Matching Framework described in Section 11, the completed framework was evaluated using controlled requirement-evidence cases.

The purpose of this evaluation was to examine whether the implemented framework could correctly identify supporting resume evidence, particularly when the wording of the job requirement and resume differed.

The results reported in this section represent controlled experimental evidence from the implemented framework. They should not be interpreted as large-scale benchmark results or as proof that CareerFit universally outperforms other resume-job matching systems.

13.1 Controlled Skill-Matching Evaluation

A separate controlled skill-matching evaluation was performed using 10 manually defined cases to verify the behavior of the supporting skill-extraction and matching logic.

The results were:

Metric	Result
Precision	1.000
Recall	1.000
F1-score	1.000
Controlled cases	10

These results show that the implemented skill-matching logic behaved as expected on the selected controlled cases.

However, the dataset is small and manually constructed. The result should therefore be interpreted as verification under the tested conditions rather than evidence of perfect skill matching on unrestricted real-world resumes and job descriptions.

13.2 Controlled Paraphrase Evidence-Ranking Comparison

The completed CareerFit Requirement-Aware Explainable Matching Framework was also evaluated using five controlled paraphrase-ranking cases.

Each case contained a job requirement, a relevant resume evidence segment, and an unrelated alternative segment. The expected supporting evidence was manually defined before the comparison.

The evaluation examined whether each method ranked the expected resume evidence correctly.

The recorded results were:

Method	Correct Evidence Rankings	Accuracy
Keyword overlap	2/5	40%
TF-IDF cosine similarity	2/5	40%
CareerFit Requirement-Aware Explainable Matching Framework	5/5	100%

Within these five controlled paraphrase cases, the completed CareerFit framework retrieved the expected supporting evidence in all five cases.

Keyword overlap retrieved the expected evidence in two out of five cases, while TF-IDF cosine similarity also retrieved the expected evidence in two out of five cases.

Within these five controlled cases, CareerFit retrieved the expected evidence more often than the two reported lexical baselines. This pattern is promising for paraphrased wording, but the small controlled sample does not support a broader effectiveness claim. For example, consider the requirement:

“Collaborate with cross-functional teams to deliver customer-facing features.”

and the resume evidence:

“Worked with product, support, and engineering groups to launch new features.”

Keyword overlap is limited because the two statements use different terminology. The CareerFit framework can assign greater importance to semantic evidence for this type of behavioral or responsibility-based requirement and retrieve the relevant resume statement.

The result should not be interpreted as meaning that CareerFit is generally 100% accurate. The evaluation contains only five controlled paraphrase cases and therefore supports a limited conclusion about the behavior of the framework under those specific experimental conditions.

13.3 Interpretation of Results

The completed evaluation provides evidence supporting several design decisions in the CareerFit framework.

First, direct lexical matching remains valuable for explicit terminology. Keyword and alias matching is transparent and can reliably identify skills, technologies, certifications, and other requirements when the same terminology is present in both the resume and job description.

Second, the controlled paraphrase experiment demonstrates a limitation of purely lexical approaches. Keyword overlap and TF-IDF each identified the expected evidence in 2 out of 5 cases, indicating that both approaches can struggle when relevant experience is expressed using substantially different vocabulary.

Third, the completed CareerFit framework identified the expected evidence in all 5 controlled paraphrase cases. This supports the design decision to combine lexical and semantic evidence at the requirement level rather than applying only a single lexical similarity measure.

The result also supports the purpose of requirement-aware weighting. Requirements such as behavioral competencies and responsibilities may benefit from greater semantic influence, while explicit skills and qualifications remain more dependent on direct evidence.

An equally important design decision is evidence grounding. CareerFit does not use semantic similarity as unrestricted proof that every requirement is satisfied. For explicit qualifications, direct or appropriately structured evidence is still required before the strongest support classification can be assigned.

For example:

Job requirement:

“AWS Certified Solutions Architect.”

Resume evidence:

“Developed cloud-based distributed applications.”

Although the two statements are semantically related, cloud-development experience does not demonstrate possession of the certification. The evidence-grounding constraint is designed to prevent this type of semantic false positive from producing an unsupported strong match.

The controlled results are consistent with the intended benefit of combining lexical and semantic evidence for paraphrased cases under the tested conditions.

However, the evidence is not sufficient to claim universal superiority. The experimental dataset is small and manually labelled, and broader evaluation across additional resumes, job domains, requirement types, and independent annotators would be required to support stronger generalization claims.

The appropriate conclusion is therefore that the completed CareerFit framework demonstrated promising and explainable matching behavior within the controlled evaluation, while larger-scale validation remains future work.

13.4 Explainability and Usability Evaluation

Explainability is evaluated separately from matching accuracy because a correct numerical score is not sufficient if the user cannot understand how the result was produced.

CareerFit is designed so that each requirement-level decision can be traced to visible resume evidence. The explainability evaluation therefore focuses on whether the system provides sufficient information for a user to understand the reason behind a classification.

The following criteria are used:

Evaluation Criterion	Expected Behavior
Requirement Visibility	Each extracted job requirement is displayed separately rather than being hidden inside an overall match score.
Evidence Visibility	For supported requirements, the system displays the resume segment selected as supporting evidence.
Classification Clarity	Each requirement is presented as Matched, Partially Matched, Weakly Supported, or Missing.
Score Interpretability	Numerical scores are accompanied by understandable classifications rather than displayed without explanation.
Gap Identification	Missing and weakly supported requirements are clearly distinguished from successfully supported requirements.
Recommendation Traceability	Recommendations can be linked to specific missing, weak, partial, or ATS-related issues.
ATS Separation	ATS preparation results are presented separately from job-content alignment so that users can distinguish structural issues from qualification gaps.

A particularly important measure is **evidence correctness**. The system should not only display an explanation; the displayed resume evidence should actually correspond to the requirement being evaluated.

For this reason, explainability is connected to the evidence-retrieval evaluation described in Section 12.3.2. If CareerFit assigns the correct support classification but selects an unrelated resume sentence as justification, the result is not considered fully explainable.

For example:

Job requirement:

“Experience developing REST APIs.”

Expected supporting evidence:

“Developed RESTful backend services using Django REST Framework.”

A strong explainable result should identify the correct requirement, return the relevant resume statement, assign an appropriate support classification, and make the connection understandable to the user.

Usability is evaluated at the workflow level by checking whether the user can move from resume input to job selection, analysis, report review, and application tracking without requiring knowledge of the underlying matching algorithms. The interface design also follows general usability principles such as visibility of system status, consistency, clear feedback, and error recovery [13].

The research focus, however, remains on **explanation quality at the requirement-evidence level**. The goal is not merely to make the interface visually understandable, but to ensure that the reasoning presented to the user corresponds to the evidence used by the matching framework.

A larger future usability study with independent participants could measure explanation usefulness, perceived clarity, trust, and task completion. Such a study is outside the scope of the current controlled evaluation.

14. Implementation Timeline

The project followed a phased implementation plan with a total planned workload of approximately 100 hours. The implementation strategy prioritized the core workflow first: resume input, job input, matching, scoring, and report generation. Optional features were addressed after the core workflow became stable.

Phase	Dates	Hours	Main Activities	Deliverable
1	Week 1	8	Review tools and finalize requirements	Problem statement, scope, requirements
2	Week 2	8	Design architecture, workflow, and evaluation strategy	Architecture and evaluation plan
3	Week 3	9	Build account, profile, and resume-input workflow	Basic user/profile/resume structure
4	Week 4	9	Implement resume parsing and preprocessing	PDF/DOCX/TXT parsing and editable preview
5	Week 5	9	Add job-description input, URL input, and limited job retrieval	Job input and search module
6	Week 6	9	Implement skill and requirement extraction	Extracted skills and requirement phrases
7	Week 7	10	Implement keyword and TF-IDF matching	Baseline matching methods
8	Week 8	10	Add mapping, scoring, ATS checks, and recommendations	Explainable report logic
9	Week 9	8	Build report page, role insights, and tracker workflow	Report UI and tracker prototype
10	Week 10	8	Add optional semantic evaluation and coaching path	Semantic comparison setup
11	Week 11	12	Testing, evaluation, final report, and presentation	Final prototype, results, report, presentation

14.1 Critical Dependencies

Task	Dependency
Resume analysis	Resume upload or pasted text must be available.
Job comparison	Resume text and job description must both be available.
Requirement extraction	Job description preprocessing must be completed.
Resume evidence mapping	Resume parsing and requirement extraction must be completed.
Score calculation	Requirement matching logic must be implemented.
Report generation	Scores, evidence categories, and ATS checks must be available.
Evaluation	Matching methods and sample cases must be prepared.

15. Risks and Mitigation

CareerFit includes both software-development risks and research-validity risks. Technical risks affect whether the prototype operates correctly, while research risks affect whether the conclusions drawn from the matching experiments are reliable.

Because the main academic contribution is the proposed requirement-aware matching framework, particular attention is given to evaluation size, semantic false positives, requirement classification, and the possibility that predefined scoring parameters influence the experimental results.

Risk ID	Risk Description	Category	Likelihood	Impact	Score	Mitigation
R1	Job-provider APIs may be unavailable or incomplete.	Data / Technical	3	4	12	Use multiple accessible sources and allow manual job-description input.
R2	Resume parsing may be inaccurate for complex PDFs.	Technical	3	3	9	Keep extracted resume text editable before analysis.
R3	Individual lexical methods may miss paraphrased resume evidence, while semantic methods may introduce false-positive matches.	Research / NLP	3	4	12	Compare lexical and semantic baselines, use requirement-aware weighting, and apply evidence-grounding constraints for explicit requirements.
R4	Project scope may become too large.	Schedule / Scope	4	4	16	Prioritize the core workflow and keep advanced features optional.
R5	Users may misunderstand scores as hiring predictions.	Usability / Ethics	2	4	8	State that scores measure alignment and readiness, not hiring probability.
R6	Optional AI coaching may be imperfect.	AI / Quality	2	3	6	Keep AI coaching optional and separate from deterministic scoring.
R7	Resume data may raise privacy concerns.	Privacy / Security	3	4	12	Use authenticated records and explicit consent for optional AI.
R8	A small controlled evaluation dataset may produce results that do not generalize to broader resume-job matching scenarios.	Research	4	4	16	Clearly state the controlled-evaluation limitation.
R9	Manually defined	Research Validity	3	3	9	Define labels before final

	requirement classifications or supporting evidence may contain subjective judgement.					evaluation, document labelling criteria clearly, and identify multiple independent annotators as future work.
R10	Predefined requirement-aware weights or classification thresholds may influence the results.	Research Design	3	4	12	Fix parameters before the primary evaluation and examine their effect through sensitivity and ablation analysis.

Overall, the identified risks are managed through fallback input options, editable resume content, modular system design, evidence-grounding rules, and clear separation between software verification and matching evaluation. The most significant remaining risks concern the limited size of the controlled evaluation dataset, subjective ground-truth labels, and predefined scoring parameters. These factors are acknowledged as limitations of the current study and restrict the conclusions to the experimental conditions reported in this project.

16. Future Work

Future work should focus primarily on strengthening the research evaluation.

A larger and more diverse dataset should be created using resumes and job descriptions from different industries, experience levels, and writing styles. This would provide stronger evidence about whether the behavior observed in the controlled cases generalizes to broader resume-job matching scenarios.

Multiple independent annotators should also be used to define expected requirement classifications and supporting resume evidence. Inter-annotator agreement could then be measured to assess the reliability of the ground truth.

Further experiments should evaluate the sensitivity of the framework to different matching weights and classification thresholds. The ablation analysis described in Section 12.5 can also be expanded to measure the contribution of requirement-aware weighting, semantic matching, skill aliases, and evidence-grounding rules.

The semantic component can be extended using embedding-based models and compared with the current deterministic semantic approach. Particular attention should be given to improving paraphrase recognition without increasing false-positive matches for explicit qualifications.

Finally, where suitable datasets and implementations are available, CareerFit could be evaluated against published person-job fit methods using common benchmark conditions.

Overall, the current evaluation should be interpreted as controlled evidence about the behavior of the proposed CareerFit framework rather than proof of universal superiority. A larger, independently labelled, and externally benchmarked evaluation would be required to support stronger general conclusions.

17. Conclusion

This project investigated how resume-job matching can be performed at the level of individual job requirements while remaining transparent to the user. CareerFit was developed as both a working application-readiness prototype and an implemented research framework for studying the strengths and limitations of different resume-job matching techniques.

The primary research contribution of the project is the **CareerFit Requirement-Aware Explainable Matching Framework**. The contribution is not the invention of keyword matching, TF-IDF, BM25, or semantic similarity individually. Instead, CareerFit organizes these complementary matching signals within a requirement-level decision process in which the type of job requirement influences how the available evidence is evaluated.

The implemented framework decomposes a job description into individual requirements and the resume into candidate evidence segments. Each requirement is assigned a broad category, candidate resume evidence is evaluated using keyword/alias, TF-IDF, BM25-style, and semantic signals, and the signals are combined using requirement-dependent weights. The framework then selects the strongest valid supporting evidence and classifies the relationship as Matched, Partially Matched, Weakly Supported, or Missing.

A central design feature is **evidence grounding**. CareerFit does not treat a high whole-document or semantic similarity score as sufficient proof that a specific requirement is satisfied. The system identifies the resume statement responsible for the decision. For explicit requirements such as certifications, degrees, or specific technologies, additional grounding rules prevent semantic similarity alone from producing an unsupported strong match.

This design addresses a limitation of applying the same matching technique uniformly to all job requirements. A behavioral requirement such as cross-functional collaboration may be expressed in a resume using substantially different wording and therefore benefit from semantic evidence. In contrast, an explicit certification such as AWS Certified Solutions Architect requires direct evidence and should not be inferred only from generally related cloud-computing experience.

The completed CareerFit framework was evaluated using controlled requirement-evidence cases. In five controlled paraphrase-ranking cases, CareerFit retrieved the expected supporting evidence in **5 out of 5 cases**, while keyword overlap and TF-IDF cosine similarity each retrieved the expected evidence in **2 out of 5 cases**. A separate controlled skill-matching evaluation using 10 cases reported precision, recall, and F1-score values of **1.000**.

The controlled results are promising. In the five paraphrase-ranking cases, CareerFit retrieved the expected supporting evidence in all five cases, while keyword overlap and TF-IDF cosine similarity each retrieved the expected evidence in two cases. Given the small, manually labelled datasets, these findings should be treated as preliminary evidence about the behavior of the framework under the tested conditions rather than as a general performance or superiority claim.

However, the results should be interpreted within the limitations of the evaluation. The datasets are small, controlled, and manually labelled. The project therefore does not claim that CareerFit is 100% accurate in general or that it universally outperforms existing resume-job matching systems. A larger and more diverse evaluation with multiple independent annotators, additional job domains, sensitivity analysis, expanded ablation experiments, and external benchmark comparisons would be required to support broader conclusions.

The web application provides the practical environment for the implemented research framework. It supports resume input and editing, job discovery, selected-job comparison, requirement-level evidence reporting, ATS-oriented preparation checks, recommendations, and application tracking. These functions are practical contributions of the prototype rather than the primary academic novelty.

Overall, CareerFit demonstrates that resume-job matching can be formulated as an **explainable requirement-evidence assessment problem** rather than only as a whole-document similarity or candidate-ranking problem. The completed framework provides a transparent method for combining different forms of lexical and semantic evidence while preserving visible justification for each requirement-level decision.

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